

Behavioral Calibration for Mathematical Reasoning under Low-Data Reinforcement Learning: Controlled Experiments and Mechanistic Evidence for 1-shot GRPO

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Abstract

This paper studies 1-shot GRPO for mathematical reasoning under an extremely low-data reinforcement-learning regime. Motivated by recent evidence that verifiable reward can remain effective with a single training example [1], it asks what kind of generation behavior changes when the data budget is so small. The analysis compares five result settings across Qwen2.5-Math-1.5B and Qwen2.5-Math-1.5B-Instruct and combines validation accuracy with diagnostics for length, confidence, entropy, and stopping behavior.

The main finding is a strong asymmetry across model families. The Base model gains roughly 0.20 validation score under meaningful reward supervision, while the Instruct model moves only slightly from an already strong starting point. Reducing the training group size from $n=8$ to $n=1$ weakens but does not remove the gain, and reward corruption causes the Base model to collapse rather than reproduce the meaningful-reward improvement. Across the successful Base settings, training also shortens outputs, lowers entropy, reduces low-confidence token mass, raises tail confidence, and makes stopping behavior sharper.

These results support a direct interpretation: in this setting, 1-shot GRPO primarily acts as a fast behavioral calibration mechanism for the Base math model. The paper contributes a matched comparison of group size and reward validity, a multi-metric account of distributional stabilization, and full-data reference results that position the low-data gains below a larger-data empirical ceiling.

Keywords: low-data reinforcement learning; mathematical reasoning; behavioral calibration; GRPO; reward corruption; token-level diagnostics

Contents

1	Introduction	1
2	Method Background and Research Questions	2
2.1	GRPO as a group-relative optimizer	2
2.2	Diagnostics beyond accuracy	2
2.3	Research questions and working hypotheses	3
3	Study Setting	3
3.1	Models and data	3
3.2	Result matrix	4
3.3	Evaluation granularity and token-diagnostic coverage	4
3.4	Full-data empirical ceiling references	5

4	Main Results	6
4.1	The broad picture	6
4.2	Base-model gains are large and repeatable	8
4.3	The Instruct model starts strong and moves little	9
5	Behavioral Evidence: Aggregate and Token-Level Views	10
5.1	Base-family behavioral changes	10
5.2	Trajectory-level dynamic summaries	13
5.3	Implications of weak regularization	16
5.4	The Instruct-family regime	16
5.5	Token-level supplementary evidence	16
5.5.1	Scope and limitations of the token-level trace set	16
5.5.2	Fixed-prompt trace evidence	17
5.5.3	Prefix-level reorganization	18
5.5.4	Local contraction from the initial to final checkpoint	19
5.5.5	Correctness-conditioned distribution separation	20
6	Group Size, Reward Signal, and Validity Risks	21
6.1	Interpreting the $n=1$ group-size setting	21
6.2	Random-reward setting	22
7	Training Dynamics and Evidence Synthesis	22
7.1	Phase distribution of gains	22
7.2	Separating the best and final checkpoints	23
7.3	Final-behavior ranking as supplementary evidence	24
7.4	Cross-metric coupling	25
7.4.1	Coupling between gains and calibration metrics	25
7.4.2	Joint frontier of final performance and stability	27
7.4.3	Role of the cross-metric figures	27
8	Discussion	28
8.1	Main supporting evidence	28
8.2	Remaining limitations	28
8.3	Practical implications	28

9 Conclusion	29
A Full-Data Reference Results	30
B Metric Reading Guide	30
C End-State Behavioral Summary for the Five Key Settings	31
D Matched n=1 versus n=8: A Compact Side-by-Side View	33
E Supplementary Comparison Among the Non-Collapse Settings	35
F Correctness-Conditioned Entropy on the Traced Subset	37
G Interpretation Boundaries for the Appendix Evidence	38

1 Introduction

Large language models have made rapid progress on complex reasoning tasks such as mathematical reasoning, code generation, and scientific question answering [3, 4, 6, 8, 9, 10, 11]. This progress is also tied to recent evaluation, process-supervision, and inference-scaling work on mathematical reasoning, including MATH-style benchmarks, Olympiad-level evaluations, process-error diagnosis, mutual-verification search, and test-time compute scaling [7, 12, 13, 17, 18, 20]. Reinforcement learning for reasoning-oriented language models has become increasingly important because many high-value tasks are not only about factual knowledge but also about behavioral control during generation. Mathematical reasoning is a representative example. A model may know the relevant concepts and still fail because it drifts into an unstable chain of thought, keeps generating after the answer is already recoverable, or assigns too much probability mass to poor intermediate decisions. From this perspective, reinforcement learning with verifiable reward is not merely a way to optimize a final answer metric; by using outcome correctness or process-quality signals, it can also reshape the trajectory by which a model organizes, calibrates, and terminates reasoning [3, 14, 15, 16, 1].

This mechanism issue became especially sharp after Wang et al. showed that reinforcement learning with verifiable reward can remain effective even with one training example [1]. In their study, a selected single example substantially improves Qwen2.5-Math-1.5B on MATH500 and across a set of mathematical reasoning benchmarks [8, 7], sometimes approaching the performance obtained from much larger RLVR datasets. Their analysis also highlights several issues that are directly relevant to this paper: the distinction between outcome reward and format correction, the role of policy-gradient optimization, the importance of exploration, and the possibility that large gains can emerge even after training accuracy on the single example has already saturated. These findings turn 1-shot RLVR from a surprising empirical anecdote into a concrete setting for studying reinforcement-learning-induced changes in reasoning behavior under extreme data scarcity.

The present paper follows that line of work but shifts the emphasis from benchmark-scale existence to controlled behavioral diagnosis. It also connects to a broader body of recent reasoning-supervision work that uses generated rationales, learned verifiers, process-level feedback, or carefully selected demonstrations to reshape model behavior [13, 14, 21, 22]. The analysis focuses on 1-shot GRPO, namely the GRPO-based instance of low-data RLVR studied here. GRPO is especially relevant in this setting because it extracts training signal from within-group comparisons instead of relying heavily on a dedicated critic [6]. As a PPO-related policy-gradient method, it is also connected to the broader reasoning-oriented post-training literature, where reward signal design, KL constraints, sample selection, and policy-update stability are central design choices [5, 3, 4, 19, 22]. For a mathematical problem, multiple samples of the same prompt often reveal rich variation:

some completions are correct and concise, some are partially correct but unstable, and some are clearly off-track. GRPO can exploit these differences to encourage better trajectories, which makes the training group size itself a meaningful variable.

Under a near-one-shot regime, the central question is not only whether the score rises, but what kind of behavior changes with it. This paper therefore asks whether 1-shot GRPO reorganizes existing reasoning behavior, whether that reorganization differs between Base and Instruct priors, and whether it depends on group-relative signal and reward validity. These questions turn a pure leaderboard-style result into an interpretable account of the model’s behavioral changes.

The rest of the paper separates those questions into method background, study setting, main results, behavioral evidence, token-level evidence, and training-dynamics synthesis. This organization keeps the experimental design in the setting section and lets the introduction close on the research questions rather than previewing every result.

2 Method Background and Research Questions

2.1 GRPO as a group-relative optimizer

GRPO is built around relative comparisons within a group of sampled responses and was introduced as a memory-efficient alternative to PPO-style optimization for mathematical reasoning models [6, 5]. In the standard configuration used here, `rollout_n=8`, meaning that each prompt contributes eight sampled candidates. This is useful for mathematical reasoning because the space of candidate solutions is structured: even when all candidates answer the same question, they can differ substantially in coherence, confidence, and stopping behavior. By comparing those candidates against one another, GRPO can push probability mass toward the better local trajectories without requiring a high-quality critic for every token prefix.

The `n=1` setting changes this central ingredient directly by using `rollout_n=1` while keeping `val_rollout_n=8` matched with `n=8`. If group-relative signal is essential, the `n=1` condition should lose part of the benefit observed in the standard setup. That makes the comparison especially informative because the main validation protocol is matched.

2.2 Diagnostics beyond accuracy

Accuracy is the primary metric, but it is too coarse to explain the effect of the training objective. This paper therefore tracks a set of validation-time diagnostics: response length, answer-tail confidence, low-confidence token ratio, mean token entropy, final EOS probability, and mean top-1 probability. If a model becomes more accurate because it is more decisive, better calibrated, and less likely to wander after reaching the right path,

these metrics should move in a coordinated way.

Under a low-data regime, this distinction becomes even more important. This paper uses *behavioral calibration* operationally to mean a coordinated movement toward shorter responses, lower token entropy, fewer low-confidence tokens, higher tail and top-1 confidence, and sharper EOS behavior while validation accuracy improves. In that definition, calibration is an observable distributional change rather than a claim that the model has learned new mathematical facts.

2.3 Research questions and working hypotheses

The paper is organized around five working hypotheses:

1. 1-shot GRPO can produce repeatable gains on mathematical reasoning, at least for a weaker Base model.
2. The gains should be much larger for the Base model than for the Instruct model because the latter starts from a stronger prior.
3. If the gains reflect calibration rather than knowledge injection, they should coincide with lower entropy, fewer low-confidence tokens, stronger tail confidence, and sharper EOS behavior.
4. A random-reward condition should not replicate the gains obtained under meaningful reward supervision.
5. If group-relative signal is essential, an $n=1$ setup should substantially weaken the gains seen under $n=8$.

The discussion section returns to these hypotheses and separates well-supported conclusions from remaining limitations.

3 Study Setting

3.1 Models and data

The comparison uses two model families: Qwen2.5-Math-1.5B and Qwen2.5-Math-1.5B-Instruct, both from the Qwen2.5 mathematical-reasoning model line [8]. Validation is conducted on MATH500, a held-out evaluation set following the MATH benchmark lineage [7]. All reported numbers are organized into a unified result table and a unified set of behavioral diagnostics.

The analyzed settings are designed to isolate a small number of variables. Most training hyperparameters remain matched across settings, including learning rate, batch sizing,

and validation cadence. The main factors are therefore model prior (Base vs. Instruct), regularization strength (standard vs. weak regularization), group size (**n=8** vs. **n=1**), and the presence or absence of meaningful reward supervision.

3.2 Result matrix

Table 1 summarizes the primary metrics for the five key settings.

Table 1: Headline results across the five key settings

Setting	initial	final	gain	best checkpoint	best score
Base n=8	0.3745	0.5733	+0.1988	156	0.5753
Base n=1 ^a	0.3630	0.5470	+0.1840	174	0.5590
Base weak-reg.	0.3745	0.5747	+0.2002	174	0.5765
Base random-reward ^b	0.3630	0.0003	-0.3628	0	0.3630
Instruct n=8	0.6837	0.6873	+0.0035	162	0.6995

^a Base **n=1** uses the same `val_rollout_n=8` protocol as Base **n=8**, so each validation checkpoint also has about 4000 records.

^b Base **random-reward** uses random wrong rewards and acts as a reward-corruption setting rather than a no-reward setting.

The table already reveals the central asymmetry of the project. The Base settings improve substantially, whereas the Instruct setting moves only slightly. It also reveals an important secondary pattern: several best checkpoints occur before the final checkpoint, indicating that the training dynamics become noisy in the later stage rather than continuing to improve monotonically.

3.3 Evaluation granularity and token-diagnostic coverage

All five settings cover validation steps from 0 to 180 with a stride of 6. Each setting uses roughly 4000 validation records per analyzed checkpoint. This matched protocol is most important for Base **n=1** and Base **random-reward**, because these settings and Base **n=8** share the same validation granularity and are therefore much more directly comparable.

Token diagnostics introduce another granularity issue. The more precise statement is that up to 64 samples per checkpoint participate in token-level diagnostic statistics, but only 8 of those samples additionally retain the full `token_diagnostics` trace. In the main **n=8** settings, those 8 complete traces usually correspond to 8 rollouts of the same fixed prompt. This is sufficient for mechanism-oriented inspection, but not sufficient for claiming full task-level token statistics over the entire benchmark. The aggregate diagnostics further confirm that the analyzed-sample count stays at 64 throughout the five settings;

the analyzed-sample fraction is about 1.6%.

3.4 Full-data empirical ceiling references

Beyond the main result table, the paper also draws on a set of full-data and format-constrained reference results. These results are not merged directly into the main result table because the training-file scope and logging coverage are not fully identical. They are nevertheless important because they provide an empirical upper bound for the low-data results. In the Base family, **full** training rises from 0.3630 to 0.6053 and clearly exceeds **1-shot** at 0.5753. In the Instruct family, **full** also exceeds **1-shot**, but the absolute gap is much smaller. At the same time, a format-constrained Base reference rises only from 0.3630 to 0.4490, remaining far weaker than the standard reward-based path.

Here, **full** follows the 1-shot RLVR paper’s data terminology and denotes training on the 1209-example DSR-sub instance pool randomly sampled from DeepScaleR-Preview-Dataset; **1-shot** denotes training on the selected example π_1 from the same DSR-sub pool [1, 2].

These reference results serve a different role from the main result table. The main table tests the claims under a matched protocol, while the full-data references position the low-data gains relative to a larger-data empirical ceiling. They therefore strengthen three claims at once: the Base family benefits much more than the Instruct family; **1-shot** training is meaningful but usually weaker than **full** training; and a weaker reward structure leads to substantially smaller gains.

Table 2: Full-data results as empirical ceiling references for 1-shot training

Model family	1-shot final	full final	final gap	best gap
Base	0.5753	0.6053	+0.0300	+0.0333
Instruct	0.6928	0.7070	+0.0143	+0.0085

Note: Gaps are computed as full-data results minus the corresponding 1-shot results. The table is used as an empirical ceiling reference, not as part of Table 1.

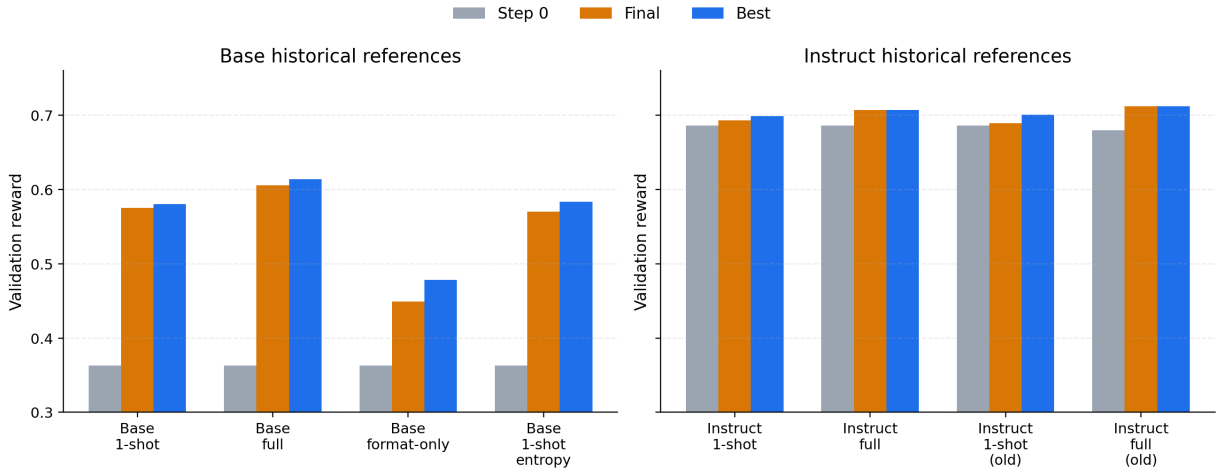


Figure 1: Initial, final, and best validation rewards for full-data and format-constrained reference results. In both model families, **full** training generally exceeds **1-shot**, while the format-constrained Base reference remains much weaker than the standard reward-based path.

Table 2 and Figure 1 therefore play a specific role in the paper: they are not sources of the main matched-protocol claims, but sources of upper-bound positioning. This distinction keeps the core claims anchored in Table 1 while still showing that the broader empirical pattern is not confined to a single result table.

4 Main Results

4.1 The broad picture

The overall picture is clear in both tabular and trajectory form: the Base family shows large gains and the Instruct family shows small gains. Figure 2 visualizes this contrast directly.

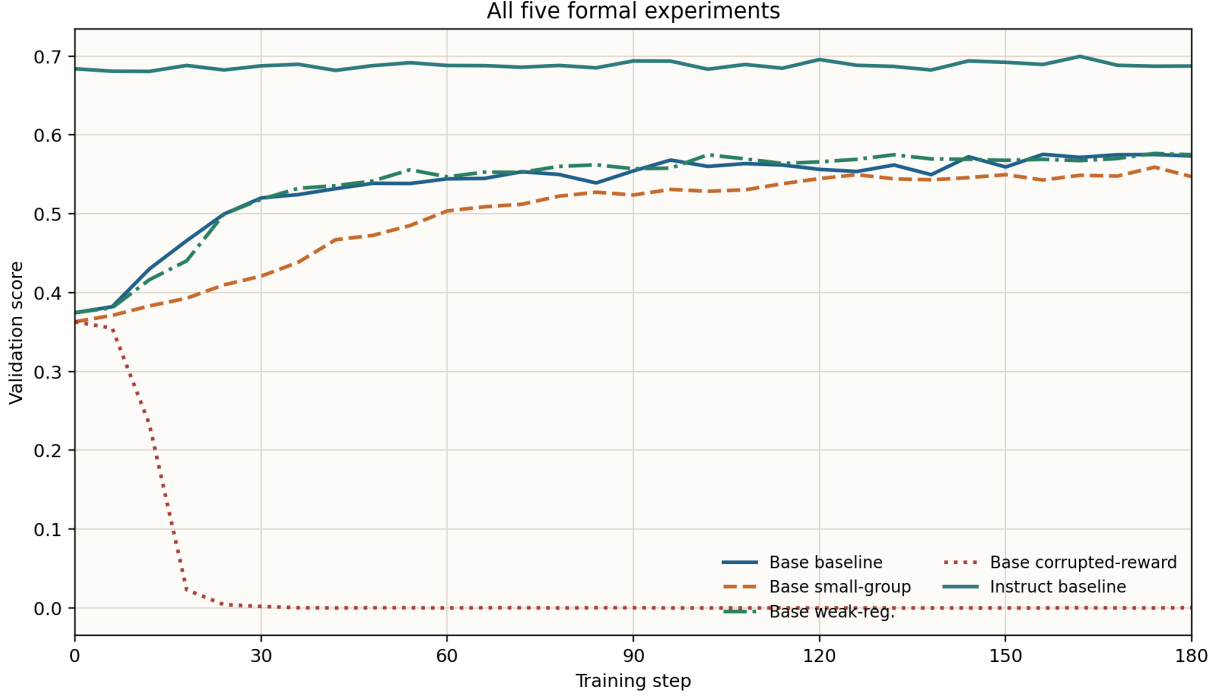


Figure 2: Validation score trajectories for all five settings. The Base settings with meaningful reward rise strongly, whereas the random-reward setting collapses and the Instruct setting remains in a high but narrow band.

Figure 2 prevents a misleading interpretation of the final table. Final values alone would obscure the fact that the Base curves undergo a genuine phase transition from low to much higher performance, while the Instruct curves mostly oscillate within an already strong range. The scale and shape of improvement are therefore different in kind, not only in magnitude.

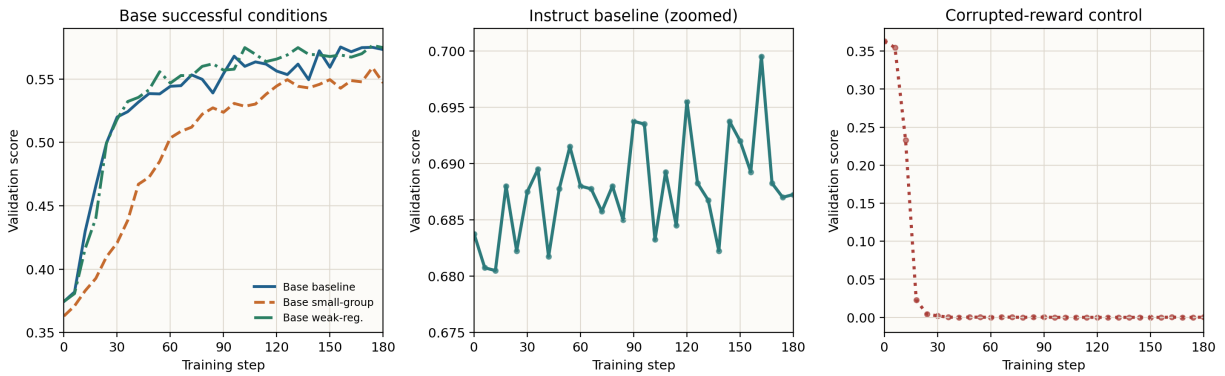


Figure 3: Zoomed validation-score views for the Base, Instruct, and random-reward settings.

Because the five settings occupy very different score ranges, the global figure is intentionally paired with a zoomed view. Figure 3 resolves two distinctions more clearly. Among the successful Base conditions, **Base weak-reg.** and **Base n=8** remain very close, whereas **Base n=1** stays consistently lower. At the same time, **Instruct n=8** can be in-

spected on its own narrow high-score band, and **Base random-reward** can be read as a distinct collapse trajectory rather than as a curve compressed against zero.

4.2 Base-model gains are large and repeatable

Base n=8 improves from 0.3745 to 0.5733. **Base weak-reg.** reaches a very similar endpoint at 0.5747, whereas **Base random-reward** collapses rather than reproducing the meaningful-reward gain. Overall, this pattern shows that the Base-family gains are not an isolated artifact of a single setting. They recur when the reward structure remains meaningful and disappear when the supervision signal is systematically broken.

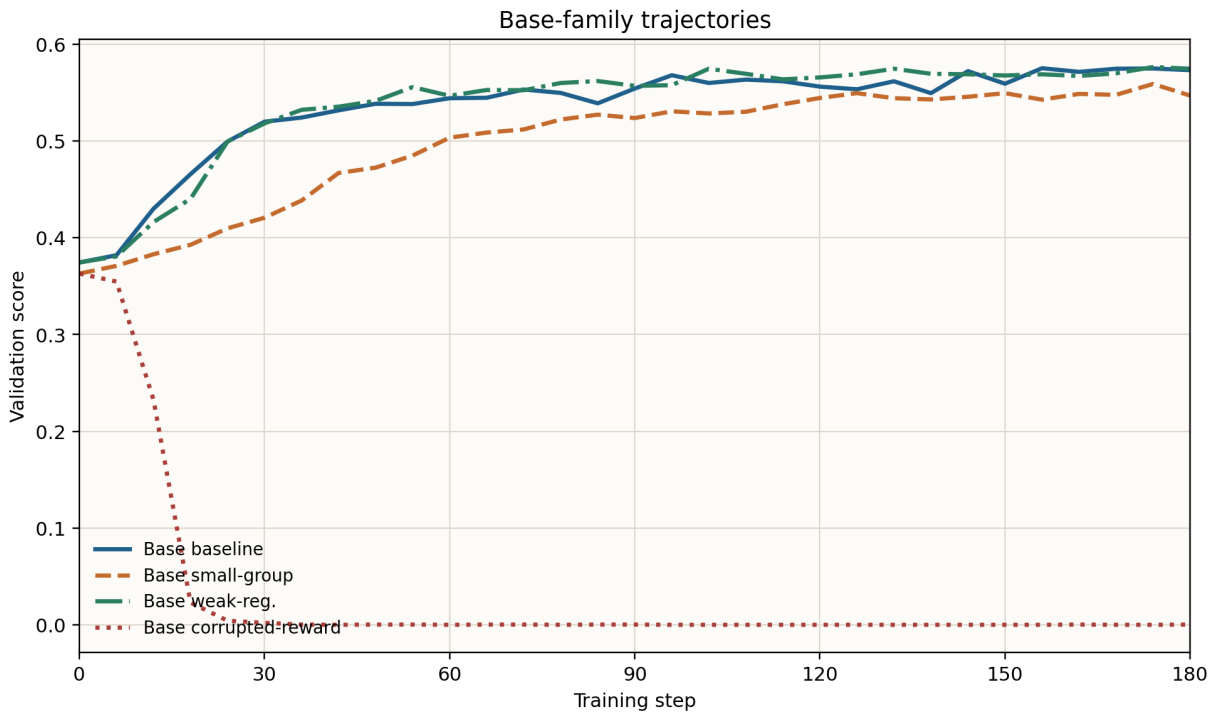


Figure 4: Base-family score trajectories. The family exhibits a pronounced early learning phase followed by higher-level fluctuation.

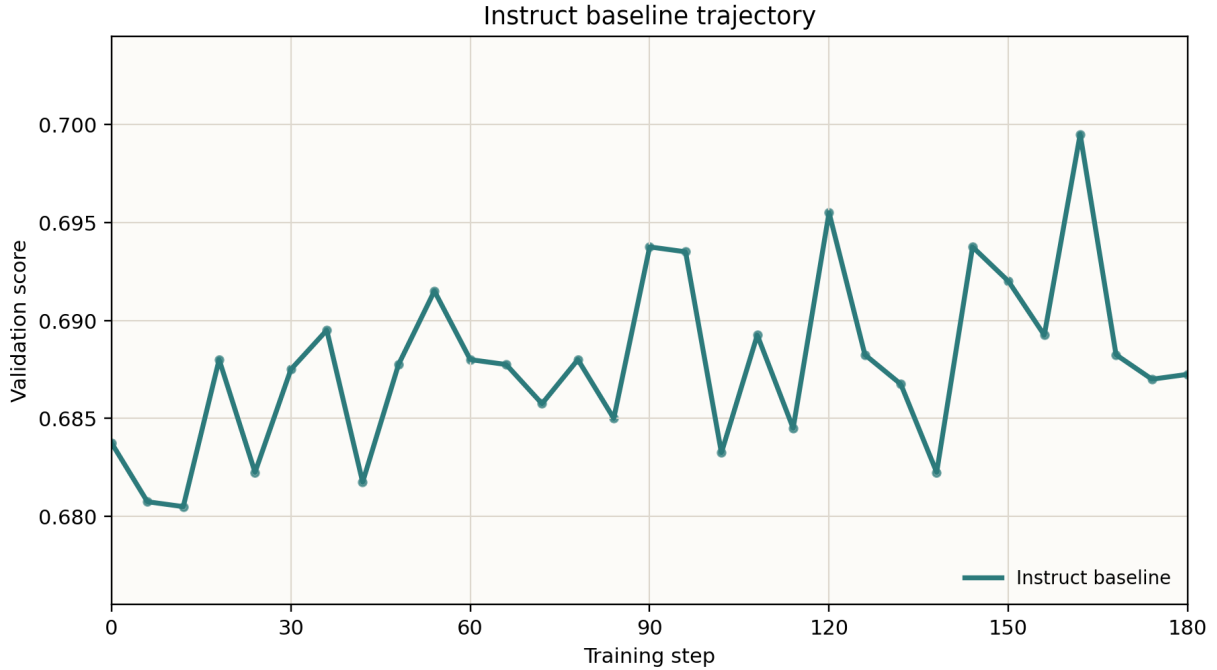


Figure 5: Instruct $n=8$ score trajectory. The curve remains in a high but narrow band throughout training.

Figures 4 and 5 also highlight an important training characteristic: the best checkpoint often arrives before the final checkpoint. Low-data RL can enter a late-stage fluctuation regime rather than improving steadily all the way to the end. For future reporting or deployment, selecting the best validated checkpoint is therefore more principled than always taking the last one.

4.3 The Instruct model starts strong and moves little

Instruct $n=8$ begins at 0.6837, far above the Base starting point. Its final value is 0.6873, with a best score of 0.6995 at step 162. These are respectable numbers, but the gain scale is tiny compared with the Base family. This pattern is consistent with an Instruct prior that already encodes much of the behavior that low-data GRPO can reasonably refine.

This difference matters for interpretation. A statement of uniform effectiveness across model families would be imprecise. A more accurate statement is that 1-shot GRPO is most useful when there is substantial behavioral slack to correct. The Base model has such slack; the Instruct model has much less.

5 Behavioral Evidence: Aggregate and Token-Level Views

5.1 Base-family behavioral changes

The Base-family gains are not just visible in score. They co-occur with a coordinated set of behavioral changes. Table 3 summarizes the final behavioral metrics for all settings, and Figures 6 through 9 show how the main Base-family metrics evolve through training.

Table 3: Final behavioral metrics for the five settings

Setting	len.	tail	low-conf	entropy	EOS	top-1
Base n=8	603.1	0.9331	0.0325	0.1839	0.9880	0.9354
Base n=1	580.4	0.9003	0.0488	0.2811	0.9215	0.9227
Base weak-reg.	578.6	0.9437	0.0294	0.1563	0.9738	0.9437
Base random-reward	1192.4	0.0001	1.0000	11.5466	0.0009	0.0021
Instruct n=8	470.3	0.9571	0.0187	0.1322	0.9926	0.9634

Note: Short labels follow the five settings in Table 1. “len.” denotes response length; “tail”, “EOS”, and “top-1” denote tail confidence, final EOS probability, and mean top-1 probability.

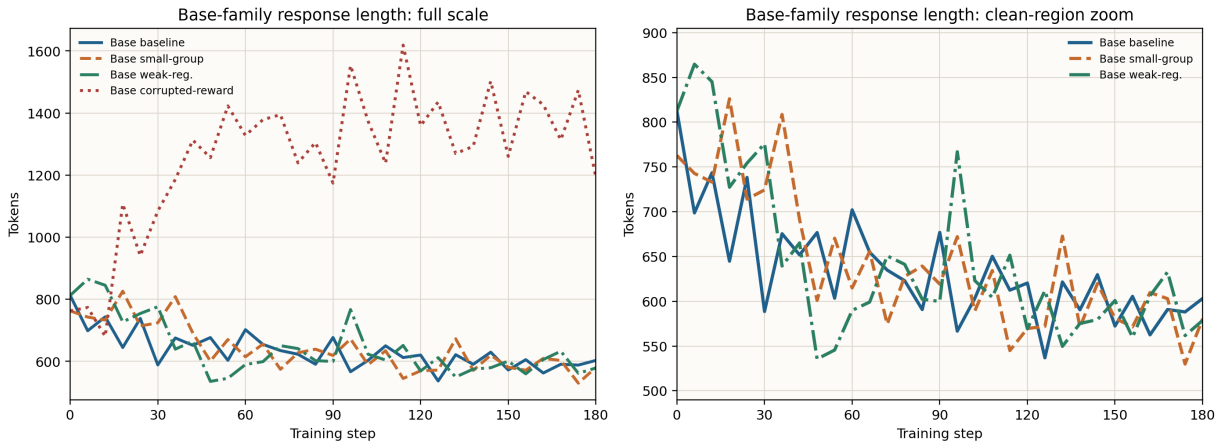


Figure 6: Validation response-length trajectories for the Base-family settings, shown with full-scale and zoomed views.

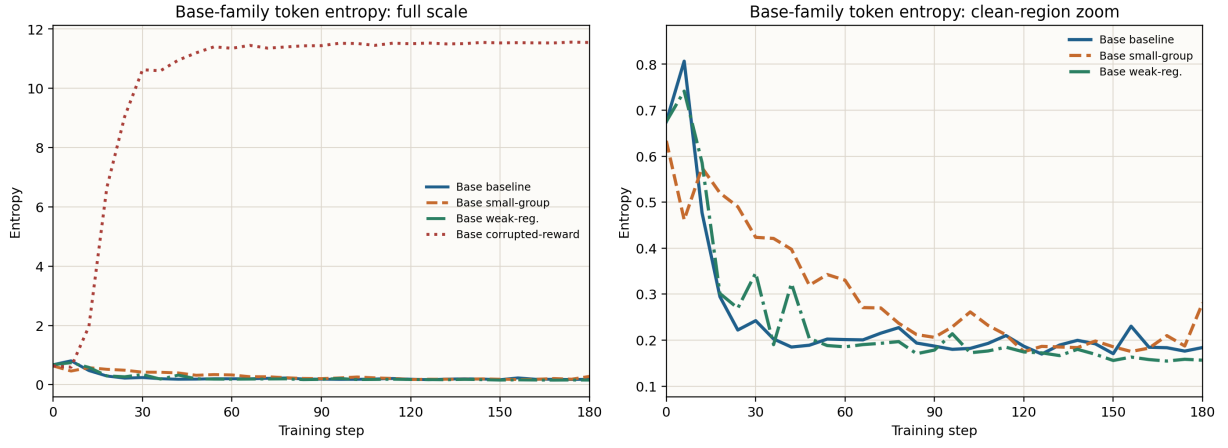


Figure 7: Validation token-entropy trajectories for the Base-family settings, shown with full-scale and zoomed views.

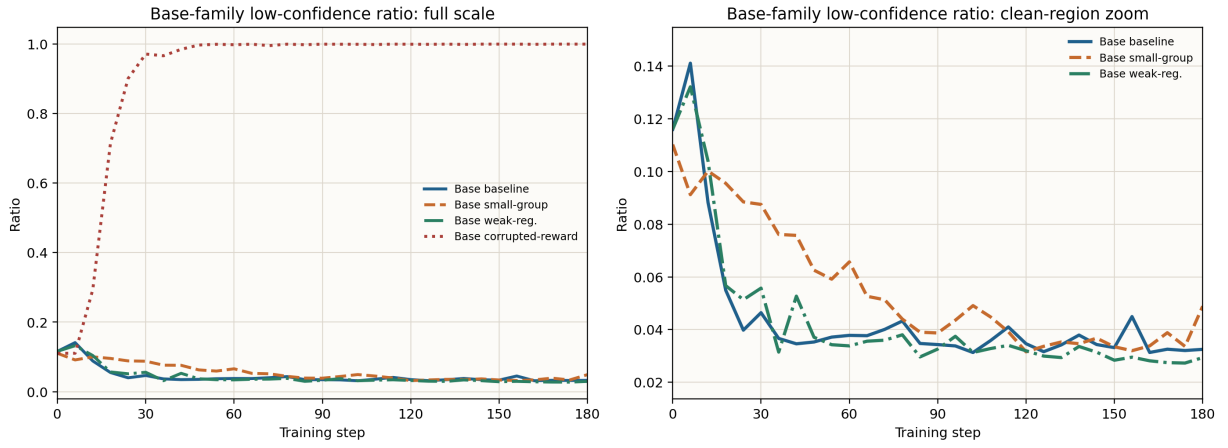


Figure 8: Low-confidence-token trajectories for the Base-family settings, shown with full-scale and zoomed views.

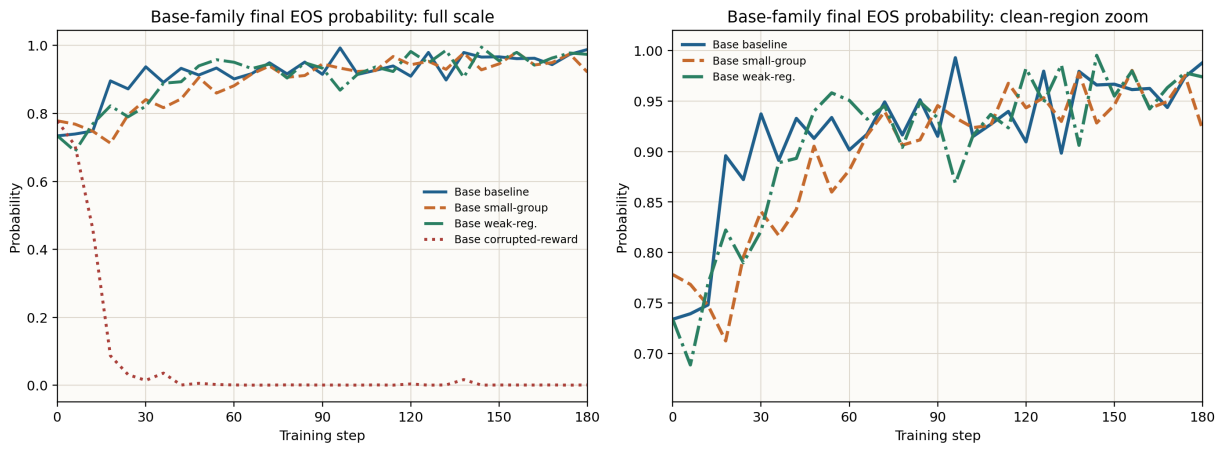


Figure 9: Final-EOS-probability trajectories for the Base-family settings, shown with full-scale and zoomed views.

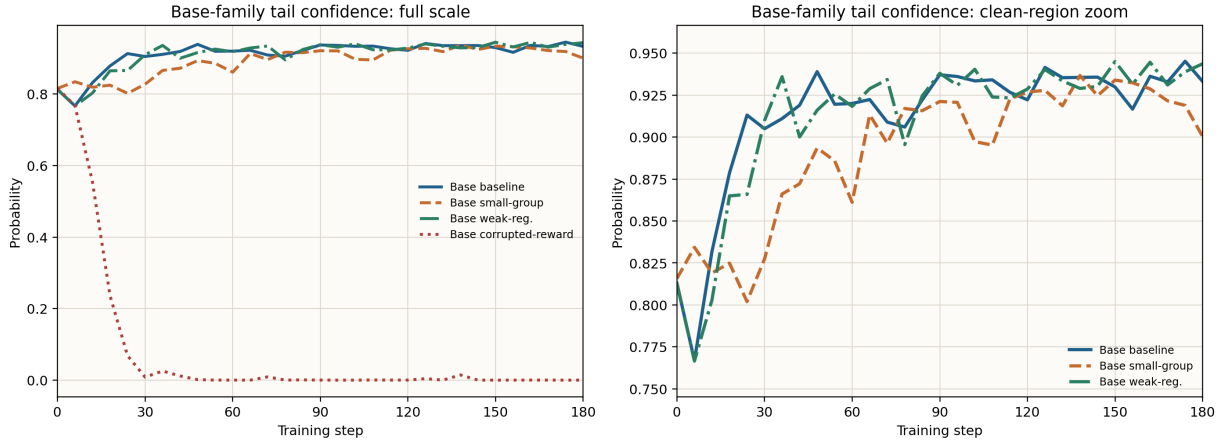


Figure 10: Tail-confidence trajectories for the Base-family settings, shown with full-scale and zoomed views.

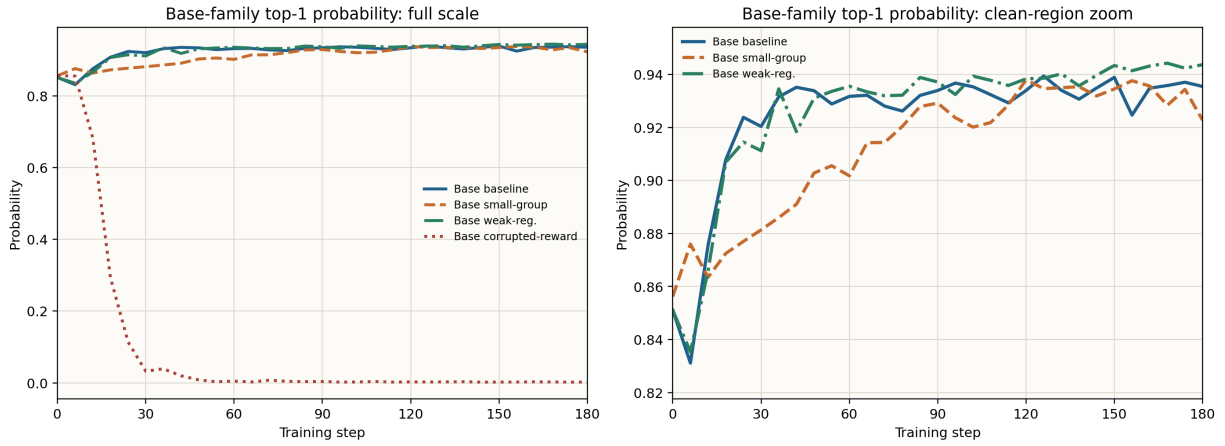


Figure 11: Mean top-1-probability trajectories for the Base-family settings, shown with full-scale and zoomed views.

Figures 6 through 11 show a consistent pattern. Each figure uses a full-scale panel plus a meaningful-reward zoom: the full-scale view preserves the abnormal direction of the random-reward setting, whereas the zoomed view prevents that outlier from compressing the differences among successful settings. Outputs get shorter, entropy falls, low-confidence tokens become rarer, and EOS probability increases, while tail confidence and top-1 probability rise. These changes are consistent with the model becoming more decisive and better calibrated. They do not indicate that performance improves through longer or more diffuse chains. Instead, the gains appear to come from a more economical and stable use of the model’s existing reasoning ability.

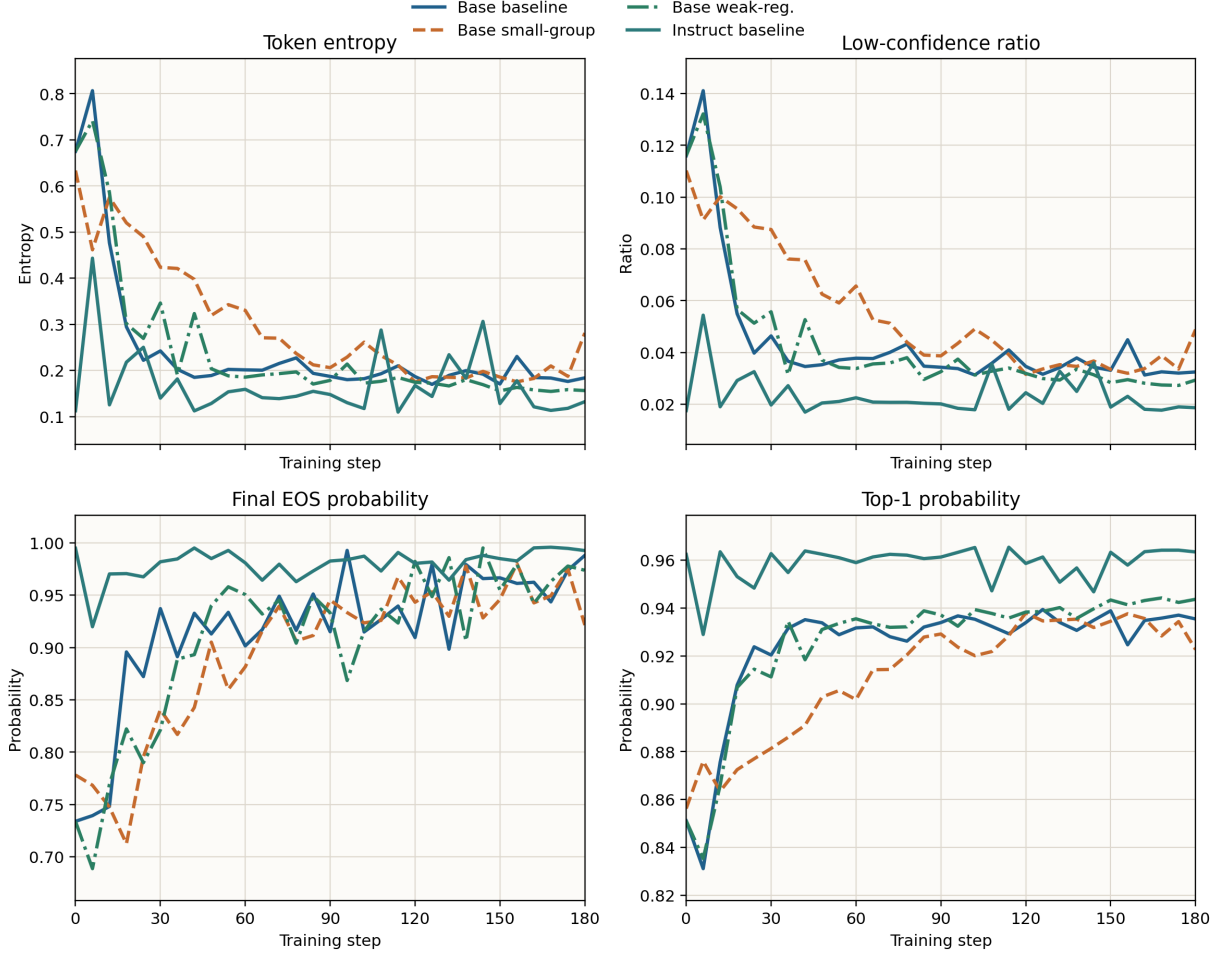


Figure 12: Fine-grained behavioral dynamics for the four non-collapse conditions.

If the random-reward setting is kept on the same axes, many of the successful-condition differences become visually compressed. Figure 12 therefore removes the collapse case and focuses on the four non-collapse conditions. The zoomed view makes a more nuanced pattern visible: **Base weak-reg.** exhibits consistently stronger contraction than **Base n=8**, **Base n=1** remains weaker on entropy- and confidence-related metrics, and **Instruct n=8** changes only modestly within an already low-entropy regime.

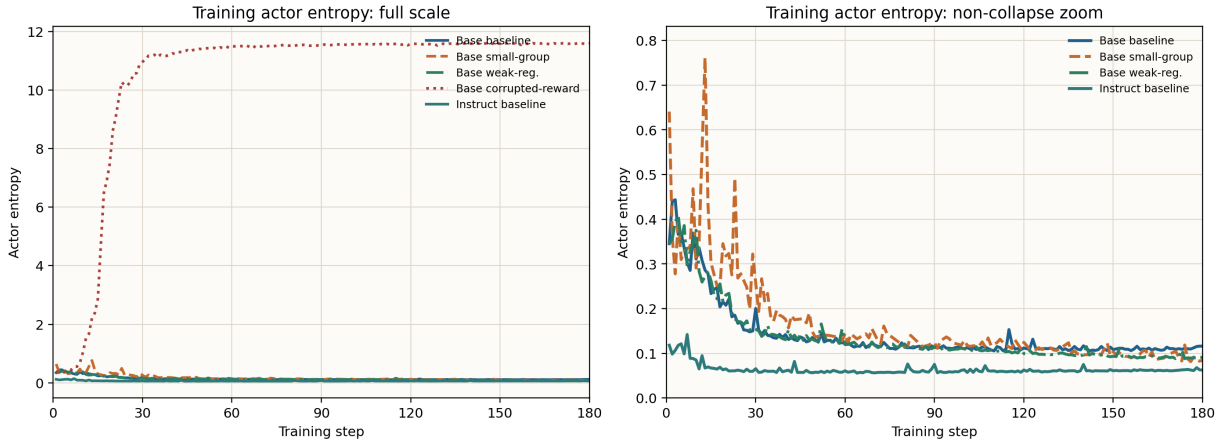
5.2 Trajectory-level dynamic summaries

The dynamic summaries sharpen this interpretation because they move the evidence from static endpoint comparison to trajectory-level mechanism analysis. They compare how training entropy, aggregated validation token entropy, low-confidence ratios, response length, and top-1 probability evolve together, rather than only which condition ends with a slightly higher reward. Table 4 summarizes the start-to-finish changes.

Table 4: Dynamic summary of training and validation diagnostics (start $\rightarrow$ final)

Setting	actor ent.	val ent.	low-conf	val len.	top-1
Base n=8	0.346 $\rightarrow$ 0.116	0.660 $\rightarrow$ 0.202	0.115 $\rightarrow$ 0.037	812.5 $\rightarrow$ 603.1	0.854 $\rightarrow$ 0.932
Base n=1	0.641 $\rightarrow$ 0.085	0.532 $\rightarrow$ 0.317	0.095 $\rightarrow$ 0.051	762.8 $\rightarrow$ 580.4	0.873 $\rightarrow$ 0.921
Base weak-reg.	0.346 $\rightarrow$ 0.091	0.660 $\rightarrow$ 0.156	0.115 $\rightarrow$ 0.030	812.5 $\rightarrow$ 578.6	0.854 $\rightarrow$ 0.944
Base random-reward	0.376 $\rightarrow$ 11.594	0.633 $\rightarrow$ 11.547	0.110 $\rightarrow$ 1.000	762.8 $\rightarrow$ 1192.4	0.856 $\rightarrow$ 0.002
Instruct n=8	0.118 $\rightarrow$ 0.062	0.111 $\rightarrow$ 0.156	0.017 $\rightarrow$ 0.022	473.2 $\rightarrow$ 470.3	0.964 $\rightarrow$ 0.961

Note: the start point is the first checkpoint with a valid value, and the end point is the final validation checkpoint. The validation token metrics come from aggregated token diagnostics with 64 analyzed samples per validation checkpoint, corresponding to about 1.6% of validation cases.

**Figure 13:** Training-time actor/entropy trajectories for the five settings, shown with full-scale and zoomed views.

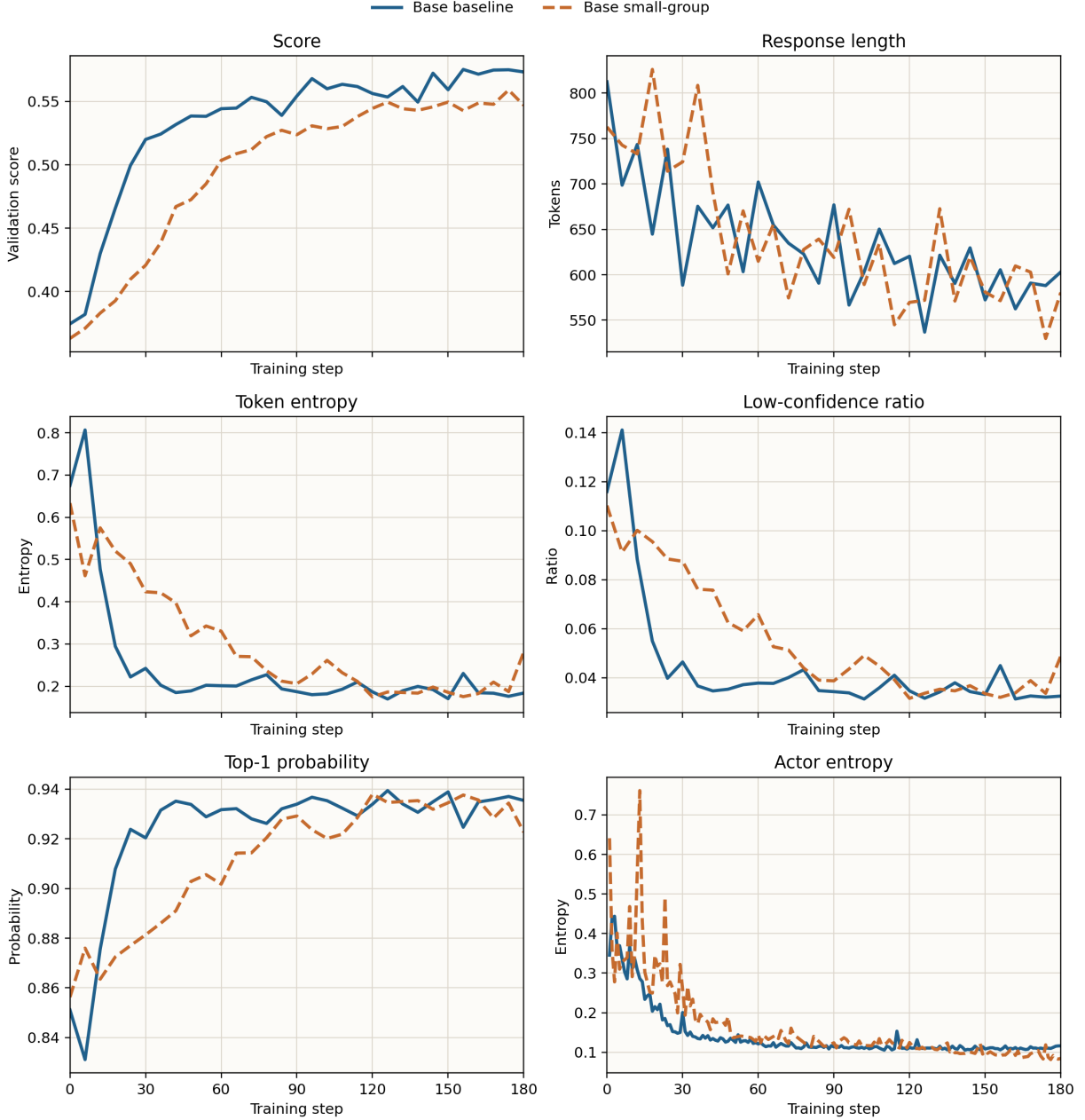


Figure 14: Matched-protocol comparison between Base $n=8$ and Base $n=1$ across six dynamics.

The strongest signal is again the asymmetry between the Base and Instruct families. For Base $n=8$, actor/entropy drops from 0.346 to 0.116, validation token entropy drops from 0.660 to 0.202, the low-confidence ratio falls from 0.115 to 0.037, aggregated top-1 probability rises from 0.854 to 0.932, and validation response length contracts from 812.5 to 603.1. Base weak-reg. is even more aggressive, ending at 0.091 actor entropy, 0.156 validation token entropy, and 0.944 top-1 probability. This is stronger evidence than score alone for the claim that the Base-family gains come from stabilization and recalibration rather than from merely producing longer or more forceful outputs.

Base $n=1$ adds a sharper and more informative nuance. Under the matched validation protocol, it improves from 0.363 to 0.547, with a best score of 0.559 at step 174. Thus,

reducing the training group size from 8 to 1 still leaves substantial learning signal, but the condition remains below **Base n=8** by about 0.026 at the final checkpoint and about 0.016 at the best checkpoint. Its final validation token entropy is 0.317, well above the 0.202 of **Base n=8** and the 0.156 of **Base weak-reg.**; its final low-confidence ratio is 0.051; and its final top-1 probability is only 0.921. Figure 14 shows that this is not a single-endpoint coincidence but a persistent gap across score, entropy, response length, and top-1 probability. Therefore, reducing group size to one does not prevent learning, but it weakens the quality of stabilization and distributional refinement.

The Instruct family shows the opposite pattern. **Instruct n=8** changes only modestly: **actor/entropy** falls from 0.118 to 0.062, validation length is nearly unchanged, and top-1 probability stays around 0.96. This is local refinement rather than structural reorganization. Once the prior is already strong, the current one-shot RL budget mainly smooths an already competent distribution instead of triggering a large behavioral transition.

5.3 Implications of weak regularization

Base weak-reg. is especially informative because it ends with a score similar to **Base n=8** while exhibiting even sharper distributional contraction in some metrics. This does not by itself make the KL and entropy terms unnecessary, but it shows that the final score alone does not capture the full training process. A condition can reach a similar endpoint while traversing a riskier or more aggressive trajectory.

This result should therefore be interpreted through both score and behavior, not through aggregate means alone. The present evidence indicates that removing those constraints did not produce a clear degradation in final accuracy; it does not show that the constraints are irrelevant.

5.4 The Instruct-family regime

The Instruct family already begins in a relatively low-entropy and high-confidence region, which explains the smaller movement in the behavioral metrics. There is no comparable transition from “messy and uncertain” to “stable and decisive,” because the starting point is already close to the latter state. The Instruct setting therefore resembles local refinement, whereas the Base settings exhibit a more substantial stabilization process.

5.5 Token-level supplementary evidence

5.5.1 Scope and limitations of the token-level trace set

The token-diagnostics analysis uses a fixed prompt and a narrower trace set. Up to 64 samples per checkpoint contribute to the diagnostic statistics, but only 8 samples retain

the full token trace, and each trace keeps only roughly the first 96 tokens. The token-level analysis should therefore be interpreted as mechanism-oriented evidence on a fixed trace set, not as a benchmark-wide token statistic.

Even with that limitation, the token-level trace set is highly valuable because it captures quantities such as token entropy, log-probability, top-1 probability, and top-k mass at the token level. These are precisely the quantities needed to test whether training is re-organizing early-token behavior. The paper therefore uses two layers of token evidence: aggregate token-summary curves, which support broad directional claims, and fixed-prompt traces, which support finer-grained mechanism inspection.

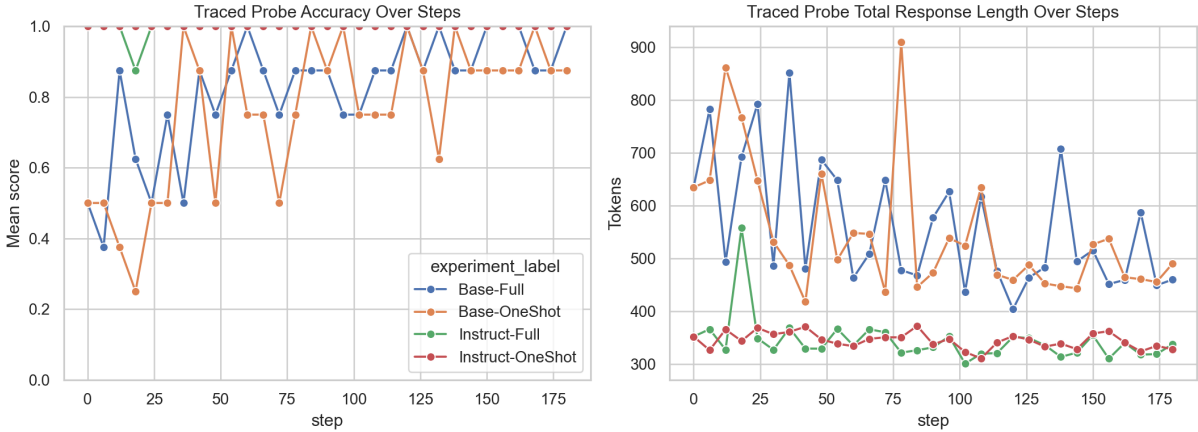


Figure 15: Fixed-prompt traced outputs across training checkpoints. The figure visualizes how prefix behavior and stopping structure evolve.

5.5.2 Fixed-prompt trace evidence

The fixed-prompt traces mirror the aggregate results. The Instruct conditions start with higher confidence and lower entropy, and their changes are comparatively small. The Base conditions undergo much larger shifts in log-probability and entropy, especially in the early token region. This provides qualitative support for the claim that Base-model training reduces prefix instability and makes the model converge on a more stable trajectory sooner.

Table 5: Final-checkpoint token-level summary on the fixed prompt

Setting	correct	traced tokens	entropy	top1 prob	top5 mass
Base-Full	correct	768	0.2036	0.9246	0.9970
Base-OneShot	correct	672	0.2005	0.9248	0.9970
Base-OneShot	wrong	96	0.2435	0.9056	0.9977
Instruct-Full	correct	768	0.0778	0.9705	0.9998
Instruct-OneShot	correct	768	0.1119	0.9580	0.9996

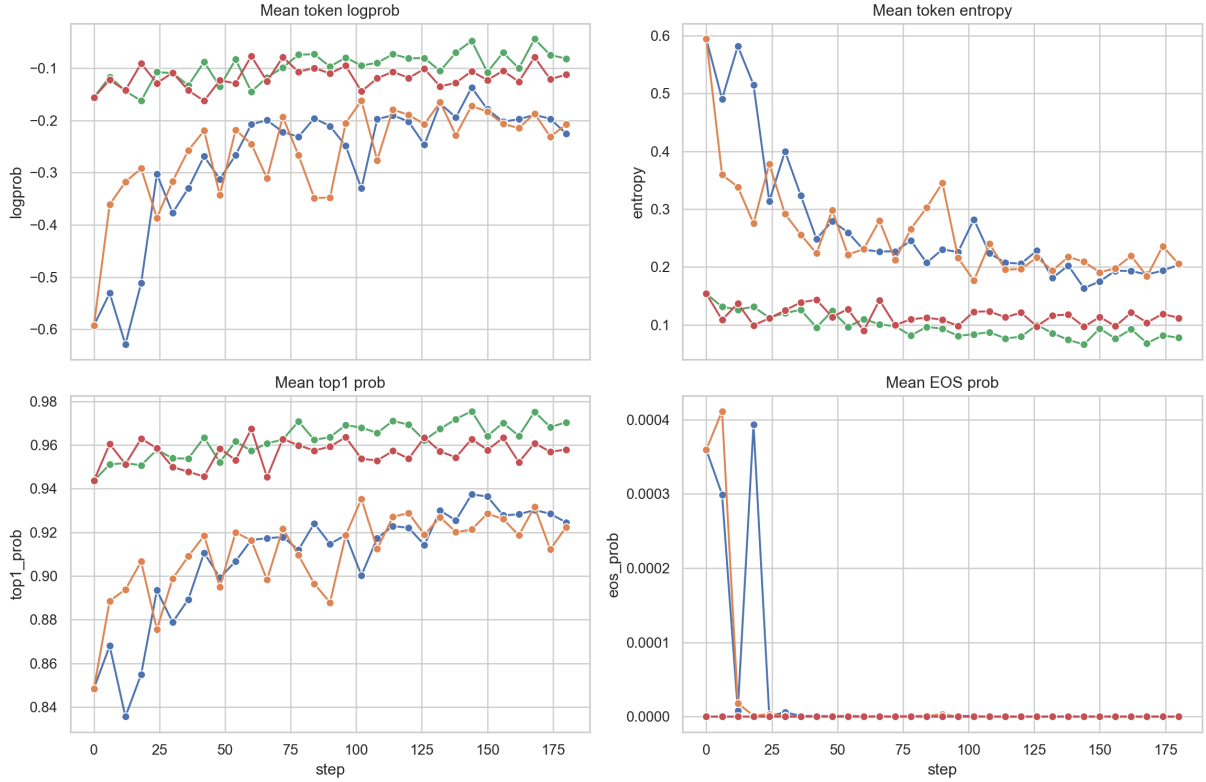


Figure 16: Token-level metrics across checkpoints. The figure is narrow in scope but informative about confidence structure and entropy reduction.

Table 6: Token-level deltas from step 0 to the final checkpoint on the fixed prompt

Setting	Δ logprob	Δ entropy	Δ top1 prob
Base-Full	+0.3673	-0.3908	+0.0762
Base-OneShot	+0.3849	-0.3885	+0.0740
Instruct-Full	+0.0744	-0.0760	+0.0268
Instruct-OneShot	+0.0435	-0.0419	+0.0143

The numbers in Tables 5 and 6 reinforce that interpretation. The Base settings exhibit substantially larger entropy decreases and larger increases in token confidence than the Instruct settings. This pattern is consistent with 1-shot GRPO acting mainly as a calibration mechanism for the weaker model family, and Figures 15 and 16 show the same pattern visually.

5.5.3 Prefix-level reorganization

The final-answer region is only part of the behavioral evidence. Figure 17 moves attention to the prefix region of the output distribution at the final checkpoint. Its main value is that it shows the training effect well before the answer is closed. For the Base

family, the more stable final outcome is not just an endpoint effect; the early trajectory already looks more structured, lower-entropy, and more concentrated. This observation is important because mathematical reasoning often fails long before the last answer token.

Figure 17 also complements the aggregate dynamic evidence. The dynamic curves show a consistent contraction over training, while the prefix plot shows that this contraction is visible positionally in the early output as well. The model is therefore not merely becoming more confident at the end; it is entering a more stable reasoning track sooner.

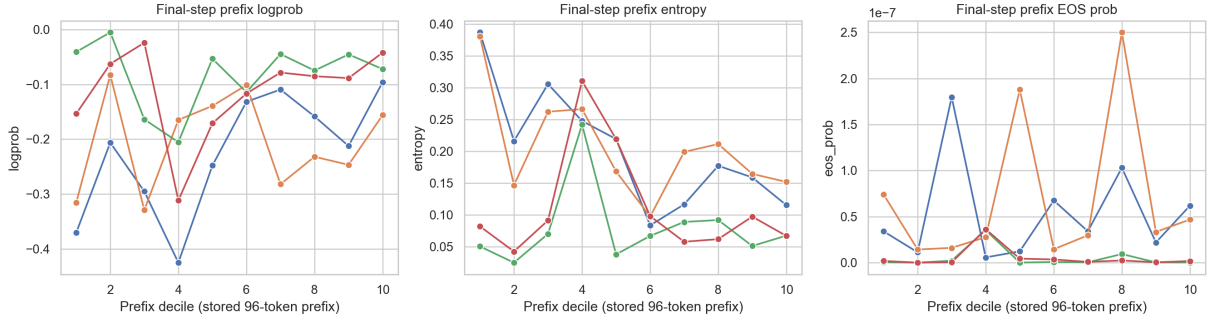


Figure 17: Prefix-region curves at the final checkpoint. The figure shows that the training effect is already visible early in the generated trajectory, not only near the final answer.

5.5.4 Local contraction from the initial to final checkpoint

Figure 18 focuses on the before-after delta itself. This makes it especially useful for distinguishing broad small improvements from substantial probability-mass redistribution in specific token regions. The figure strongly favors the latter interpretation for the Base conditions. The changes in log-probability and entropy are much larger than in the Instruct conditions, especially in the early and middle token regions.

This also helps frame **Base n=1** more carefully. The fixed-prompt trace set was not built specifically for that setting, so it is not a direct visualization of that setting. However, the mode of change it reveals matches the aggregate pattern: Base-model training can produce more concentrated token distributions, and reducing group size preserves that process only partially rather than fully.

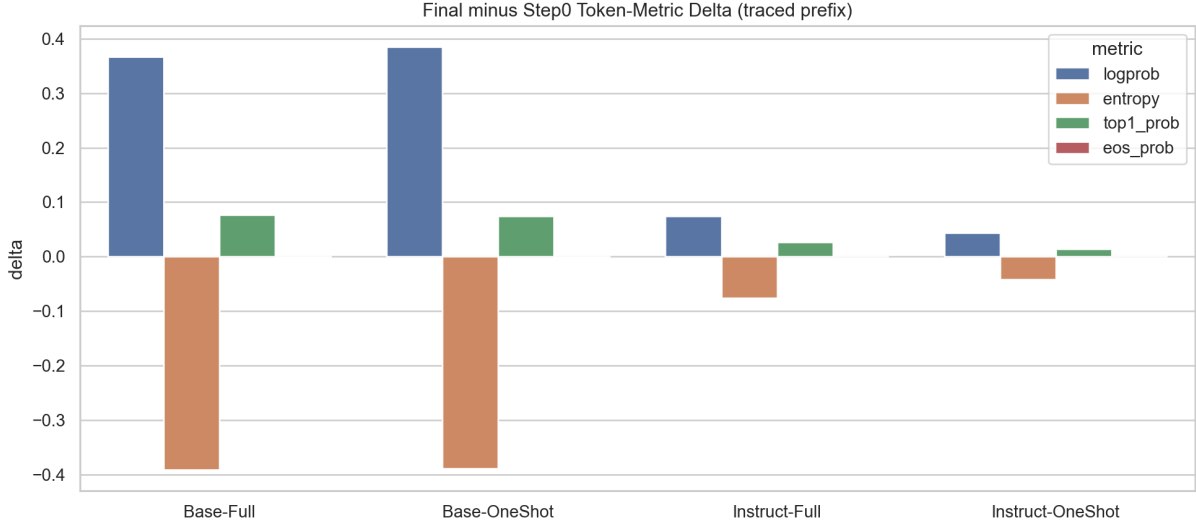


Figure 18: Token-level deltas from the initial to final checkpoint.

5.5.5 Correctness-conditioned distribution separation

To test whether distributional contraction is genuinely related to task success, we group the validation records with retained token-diagnostics summaries by response correctness and compare mean token entropy with mean top-1 probability. Figure 19 summarizes the final-checkpoint 64-sample diagnostics from the four non-collapse settings and tracks **Base** $n=1$ through training. The main result is consistent: correct responses systematically occupy the more contracted side of the distribution, with lower entropy and higher top-1 concentration.

This separation is visible across all four non-collapse conditions at the final checkpoint. For **Base** $n=8$, the mean token entropy of the correct and wrong groups is 0.165 and 0.237; for **Base** $n=1$, 0.159 and 0.570; for **Base weak-reg.**, 0.152 and 0.172; and for **Instruct** $n=8$, 0.087 and 0.499. The corresponding mean top-1 probabilities are 0.939/0.925, 0.941/0.878, 0.945/0.940, and 0.968/0.925. The magnitude of the separation varies by condition, but its direction is unchanged. The contraction and confidence concentration documented throughout the paper are therefore not merely cosmetic properties of the token distribution; they are stably associated with successful problem solving.

The checkpoint-wise grouped statistics further show that this relationship is not confined to the terminal checkpoint. In **Base** $n=1$, the correct and wrong groups already differ at **step 0**, with mean token entropy 0.353 versus 0.880, and this pattern persists to the final checkpoint, where the corresponding values are 0.159 and 0.570. Over the same interval, mean top-1 probability moves from 0.895/0.822 at **step 0** to 0.941/0.878 at the final checkpoint. In other words, the correctness-conditioned distribution gap persists throughout training rather than emerging only in a small traced subset at the end. **Base random-reward** is excluded from this figure because it has no correct responses at the final

checkpoint.

This section answers the mechanism question at two levels. Aggregate behavioral metrics show the broad stabilization pattern, while token-level traces show that the same change is visible inside generated trajectories. The evidence therefore supports the calibration account without requiring the reader to treat each diagnostic as an isolated result.



Figure 19: Correctness-conditioned token-distribution statistics. The top row compares final-checkpoint correct and wrong responses; the bottom row tracks the same grouped statistics over training for Base $n=1$.

6 Group Size, Reward Signal, and Validity Risks

6.1 Interpreting the $n=1$ group-size setting

Base $n=1$ yields a matched-protocol comparison. It improves from 0.3630 to 0.5470, with a best score of 0.5590, which demonstrates that reducing the group size to 1 does not eliminate learning.

At the same time, the matched comparison directly argues against the claim that group size is irrelevant. Under the same `val_rollout_n=8` protocol, $n=1$ is consistently weaker than $n=8$. The resulting conclusion is that group-relative signal is not necessary for gains to exist, but it appears important for reaching the best and most stable final regime.

The remaining caution is that the comparison is still single-seed rather than multi-seed.

6.2 Random-reward setting

The random-reward setting addresses a central causal issue: whether the gains are driven by meaningful supervision, or whether a randomized reward signal can produce similar gains. It is particularly informative because it directly tests whether the meaningful-reward improvement depends on the integrity of the reward signal.

Table 7: Endpoint behavior of the random-reward setting

Metric	Value
final score	0.00025
final response length	1192.4
final tail confidence	0.0001
final low-confidence ratio	1.0000
final token entropy	11.5466
final EOS probability	0.0009
final top-1 probability	0.0021

Trained under random wrong rewards, **Base random-reward** collapses rather than merely failing to improve. It begins at 0.3630 but falls to 0.00025 by the end of training; response length grows from 762.8 to 1192.4, low-confidence mass saturates to 1.0000, token entropy rises to 11.5466, final EOS probability falls to 0.0009, and mean top-1 probability collapses to 0.0021. Table 7 summarizes this endpoint regime. This result shows that the meaningful-reward gains are not reproduced when supervision is actively corrupted. It should not be read as a pure no-signal ablation: the setting injects wrong reward information, making it an intentionally severe reward-corruption control.

7 Training Dynamics and Evidence Synthesis

7.1 Phase distribution of gains

The final score does not reveal when the gains are realized. Figure 20 decomposes the score change of each setting into three coarse phases: $0 \rightarrow 36$, $36 \rightarrow 96$, and $96 \rightarrow 180$. The decomposition is intentionally simple, but it is sufficient to distinguish fast early correction from mid-stage consolidation and late-stage fluctuation.

The pattern is consistent with the paper’s main interpretation. In the Base family, most of the improvement arrives in the first two phases, especially in the rapid $0 \rightarrow 36$

stage. This is consistent with a calibration interpretation: the model is first moved out of its unstable initial regime, then consolidated, and only later enters a more variable high-score region. The Instruct family, by contrast, shows little meaningful phase-wise gain, which reinforces the claim that this setting behaves more like local refinement than like structural reorganization.

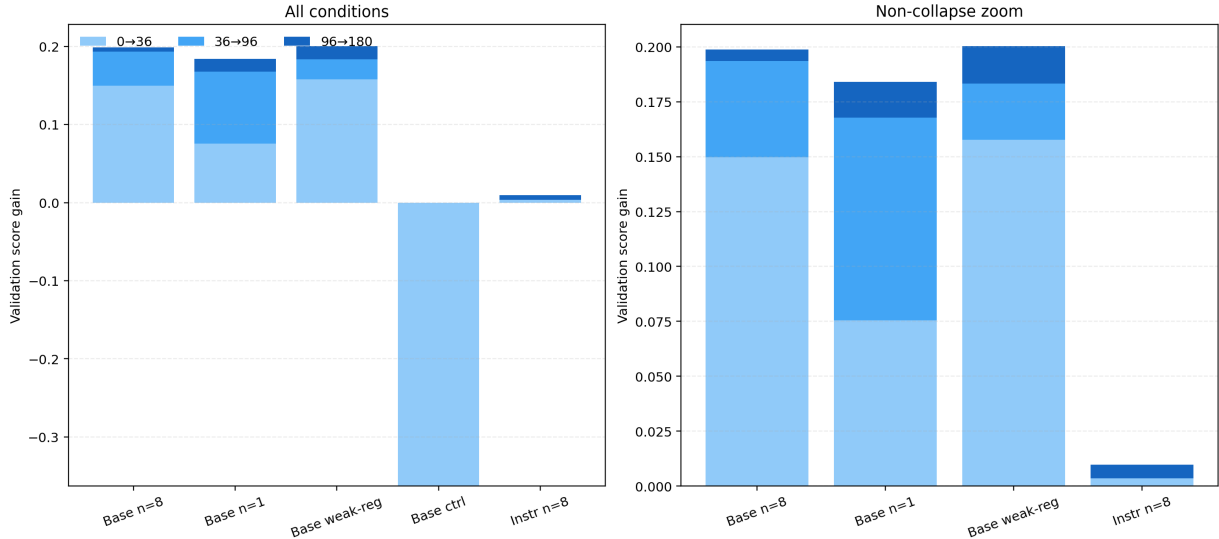


Figure 20: Stage-wise decomposition of score gains across the five settings.

7.2 Separating the best and final checkpoints

Under a low-data RL regime, the best checkpoint and the last checkpoint are often not the same. Figure 21 makes this concrete by showing both the **best score** - **final score** gap and the step at which the best checkpoint appears. This consolidates an observation that would otherwise remain scattered across the tables: several settings reach their strongest validation point before the end of training.

This distinction matters for interpretation. The training process looks less like a monotonic climb and more like a transition into a higher but somewhat fluctuating regime. For the Base settings, checkpoint selection and early stopping are not secondary engineering details but part of the optimization behavior. For the Instruct setting, once the model is already strong, the later part of training is more likely to manifest as noise than as a large deterministic gain.

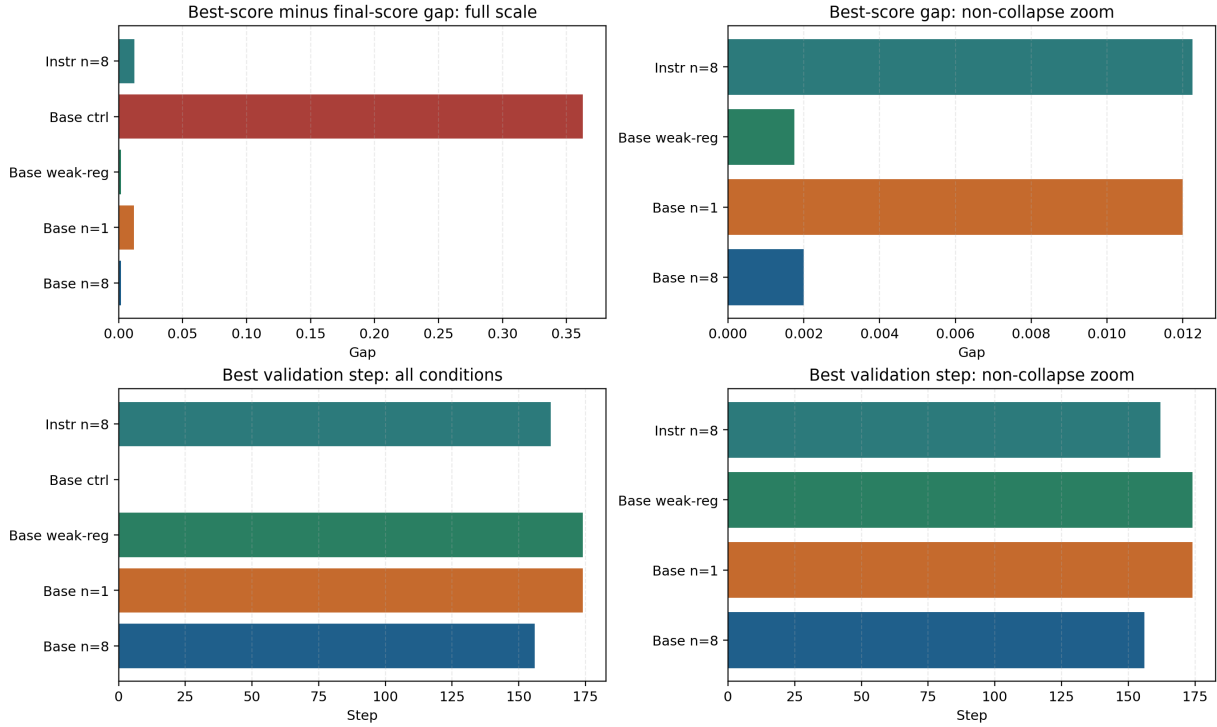


Figure 21: Comparison between the best and final checkpoints across the five settings.

7.3 Final-behavior ranking as supplementary evidence

Figure 22 ranks the five settings by several final behavioral metrics: token entropy, low-confidence ratio, top-1 probability, and response length. Compared with reporting these numbers separately, the ranking view makes the global structure easier to see. It shows, for example, that the Base-family meaningful-reward settings can end with similar rewards while still exhibiting noticeably different convergence styles. **Base weak-reg.** shows stronger contraction, **Base n=8** is more moderate, **Base n=1** remains visibly worse than **n=8** on final entropy and low-confidence mass, and **Base random-reward** lies far outside the convergence region of the non-collapse settings.

The paper should therefore not rely on reward alone. Several Base-family settings cluster very closely in final score, but the ranking plot shows that their endpoint distributions are not interchangeable. The same is true in a different way for the Instruct family: small reward differences can coexist with visibly different token-entropy structure. Together, these observations strengthen the broader methodological point of the paper: the strongest evidence comes from reading score and behavior together rather than reducing everything to a single summary metric.

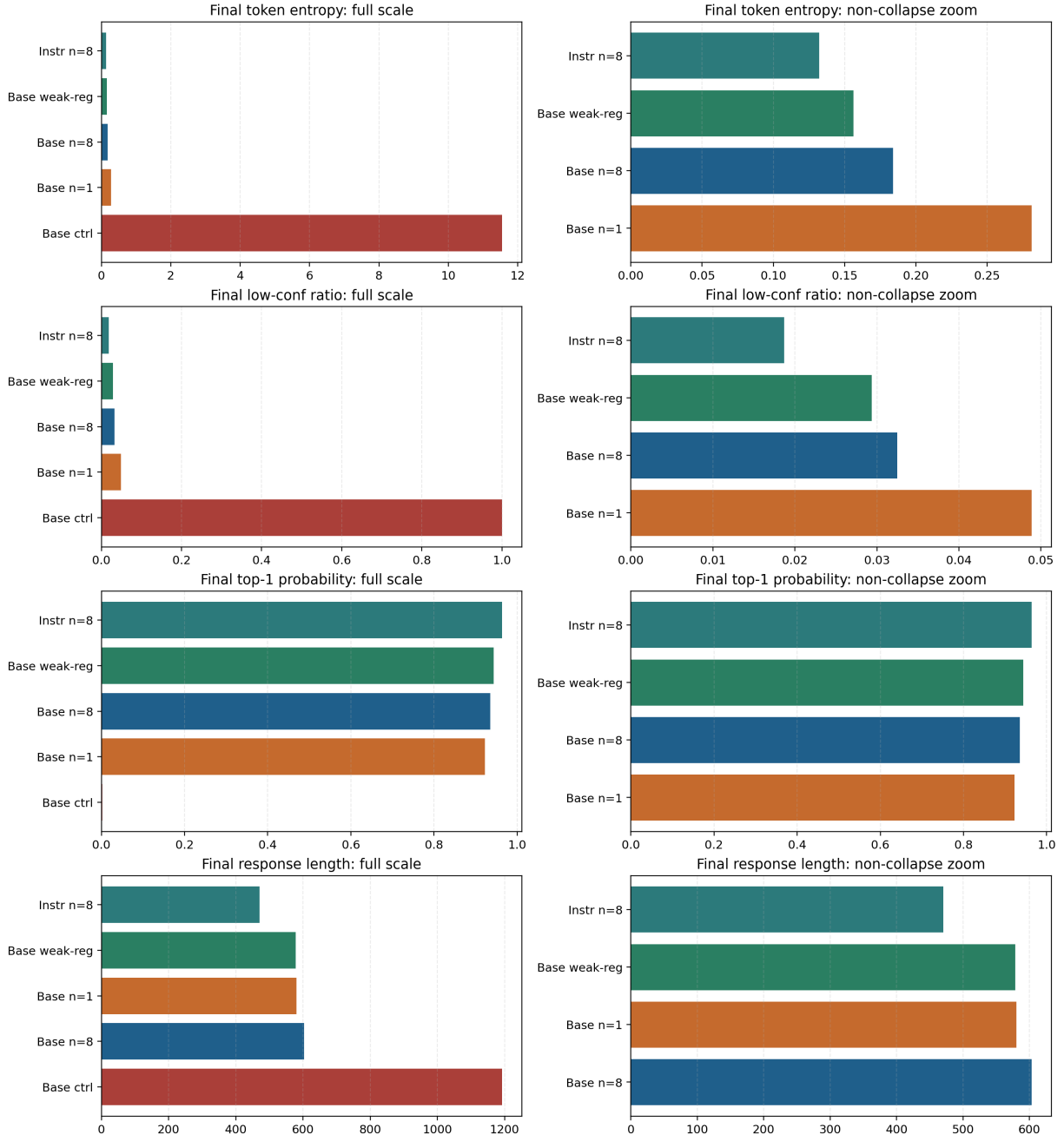


Figure 22: Final-behavior ranking across the five settings.

7.4 Cross-metric coupling

7.4.1 Coupling between gains and calibration metrics

The preceding sections separately discuss score trajectories, behavioral endpoints, and token-level supplementary evidence. A stricter test is whether those observations only point in the same direction, or whether they also co-vary at the condition level. Figure 23 addresses this issue by plotting the score gain of each setting against four dynamic change magnitudes: the reduction in training-time actor entropy, the reduction in validation to-

ken entropy, the reduction in low-confidence token ratio, and the reduction in validation response length.

Figure 23 turns the calibration interpretation into a tighter empirical pattern. The Base-family settings are not only the settings with larger score gains; they are also the settings with larger entropy contraction, larger removal of low-confidence mass, and larger shortening of outputs. **Instruct n=8** occupies a different region altogether: it improves only slightly and exhibits far smaller contraction. This coupling is more informative than listing endpoint numbers separately, because it shows that the gain and the stabilization process appear together in the same settings.

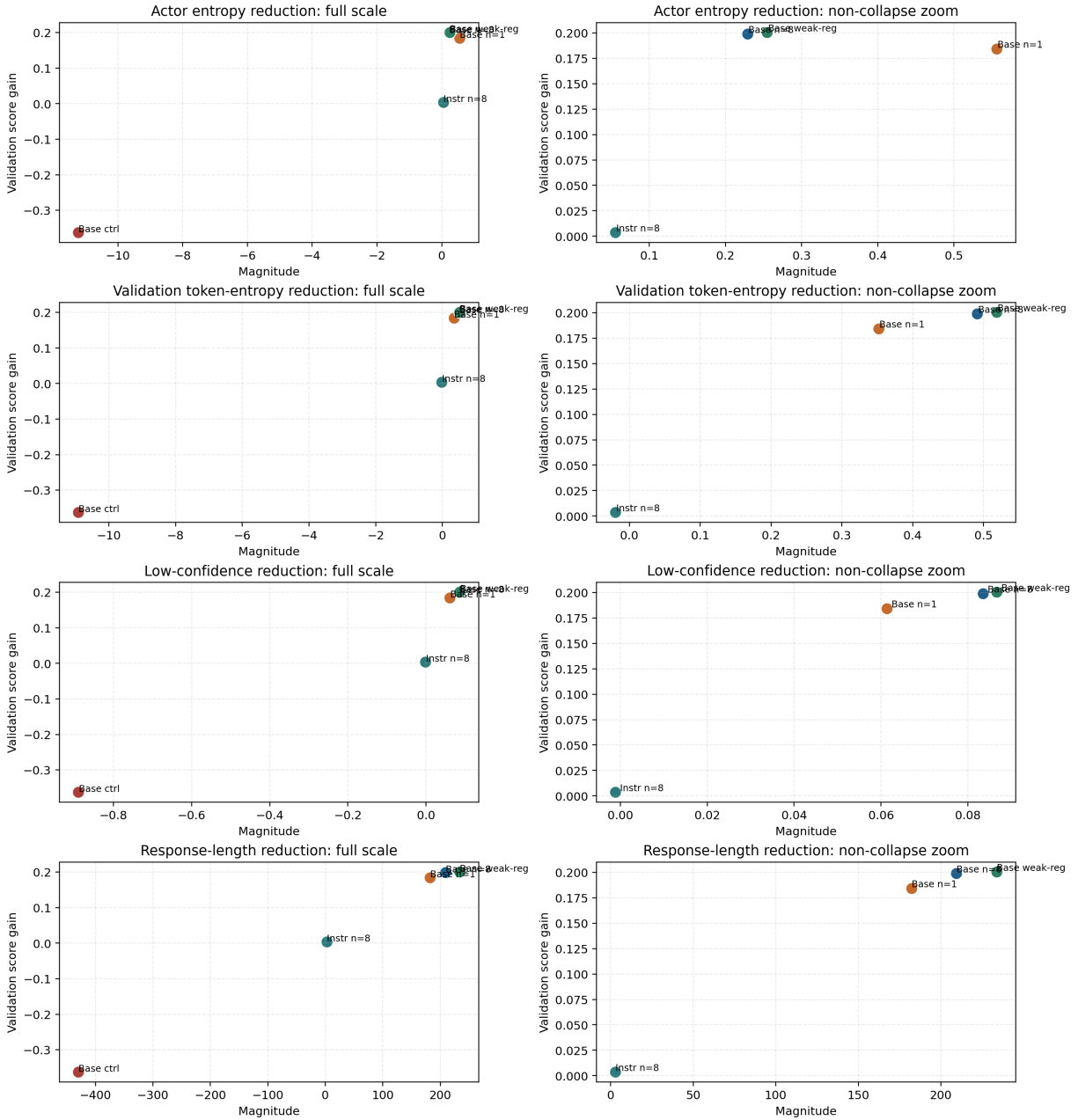


Figure 23: Coupling between validation score gain and four key dynamic changes.

7.4.2 Joint frontier of final performance and stability

It is also useful to ask whether the settings with better final reward are the same settings that end in more stable output distributions. Figure 24 places final validation score in a shared coordinate system with two endpoint stability metrics: final token entropy and final low-confidence ratio. Even with only five settings, this frontier-style view is informative because it compresses “better performance” and “more stable distribution” into the same visual frame.

Within the Base family, the trend is fairly clear: the higher-scoring settings also tend to have lower entropy and lower low-confidence mass. **Base n=1** remains visibly behind **Base n=8** on both axes, so its weakness is not confined to a single reward curve; it is reflected in the endpoint distribution itself. The Instruct family occupies a different region: the final scores are higher, but the internal spread is small, which is consistent with a model that already starts in a relatively stable regime.

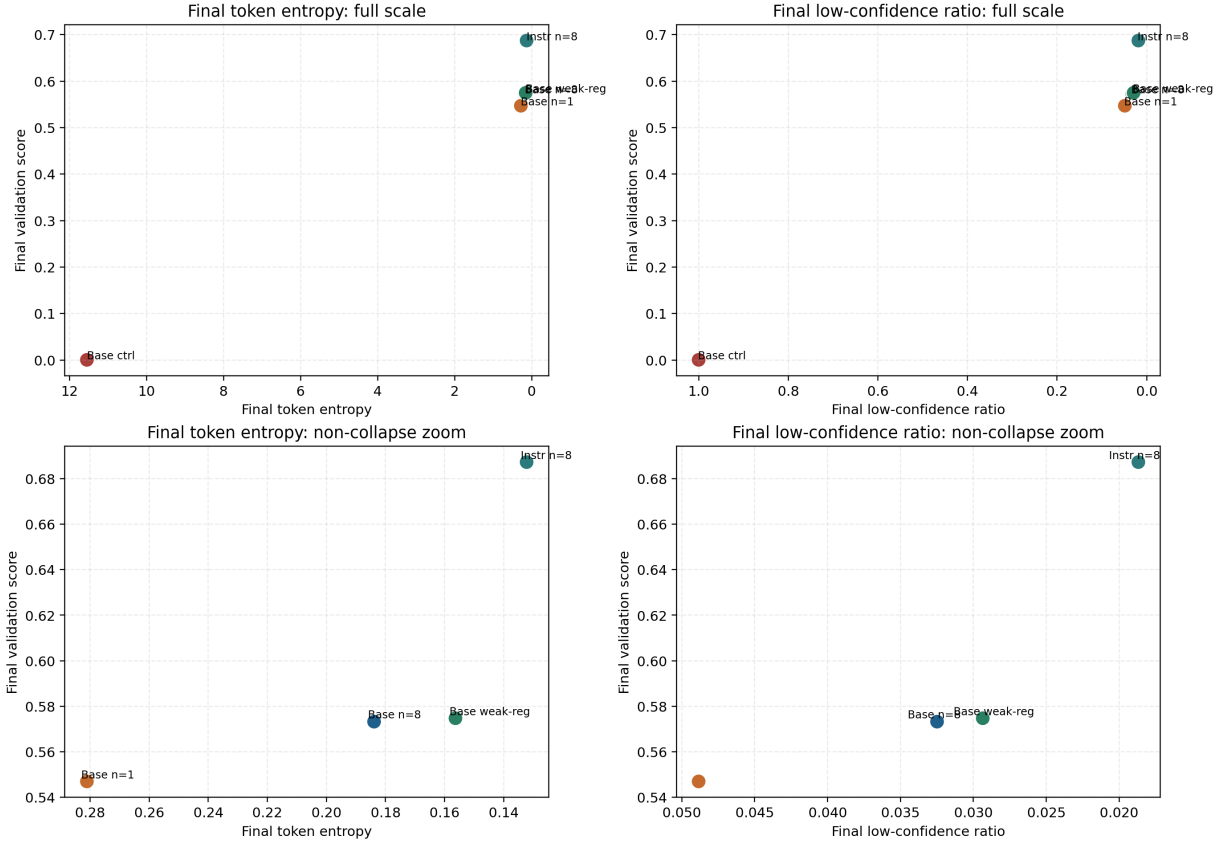


Figure 24: Joint frontier view of final score and final stability metrics.

7.4.3 Role of the cross-metric figures

Together with the preceding sections, these figures strengthen the overall evidential structure. The first layer of evidence is the score improvement itself and the distinction between best and final checkpoints. The second layer is the dynamic change in behavior:

length, entropy, top-1 probability, EOS behavior, and low-confidence mass. The third layer, added here, is the cross-metric coupling view, which puts those quantities back at the condition level and shows that they move together. This section therefore answers the timing-and-synthesis question: the gain appears early, enters a fluctuating high-score regime, and remains coupled with distributional stabilization across settings.

8 Discussion

8.1 Main supporting evidence

The strongest part of the evidence is the coherence of the Base-model pattern. Score rises sharply. Response length drops. Tail confidence rises. Low-confidence tokens become rarer. Entropy falls. EOS behavior becomes sharper. The token-level fixed-prompt traces independently show more stable early-token organization, and the full-data reference results point in the same direction. Read together, these observations make behavioral calibration the best-supported explanation under the current scope.

8.2 Remaining limitations

Four issues remain unresolved. First, the group-size result needs replication across multiple seeds. Second, the random-reward setting has not yet been matched by an Instruct-family counterpart. Third, the current diagnostics support a calibration account, but they do not fully separate calibration from knowledge acquisition across problem domains or difficulty levels. A stronger follow-up would stratify held-out problems by difficulty or topic and test whether gains are uniform, as expected under broad calibration, or concentrated in specific knowledge regions. Fourth, the full-data references are empirical upper-bound references rather than fully matched comparison items with the same diagnostic coverage.

8.3 Practical implications

The practical implication is clear: under severe data constraints, 1-shot GRPO can still be worthwhile for improving a weaker math reasoning model. The gains are substantive, and they appear to come from making the model more decisive, shorter, and better calibrated. For a model with a strong instruction-tuned prior, expectations should be more modest. In that regime, RL is closer to local refinement than to a large performance transition.

9 Conclusion

This paper characterizes the effect of 1-shot GRPO on mathematical reasoning using five key settings, dynamic behavioral diagnostics, token-level supplementary evidence, and a set of full-data references used as empirical upper bounds. Across these sources, the overall picture is consistent. The Base model benefits substantially, the Instruct model changes only slightly, and the successful Base settings exhibit a coordinated transition toward shorter, lower-entropy, higher-confidence, and more decisive outputs. The full-data references further clarify that 1-shot training captures a meaningful part of the attainable gain but should not be treated as equivalent to training with a larger data budget.

Within that evidence framework, three conclusions are the most defensible. First, 1-shot GRPO is effective for the Base math model under the present low-data regime. Second, the gain is better explained as rapid calibration than as the injection of additional mathematical knowledge, while a difficulty- or topic-stratified evaluation would be needed to fully separate those mechanisms. Third, both group-relative signal and reward integrity matter for the final regime: reducing the training group size from 8 to 1 weakens but does not remove the gain, whereas corrupting the reward signal causes the model to collapse in both score and behavioral stability. The remaining caveats are equally clear. The $n=1$ versus $n=8$ comparison is still single-seed rather than multi-seed, and the random-reward setting has not yet been matched by an Instruct-family counterpart. Within those limits, the current evidence supports interpreting 1-shot GRPO primarily as a low-data stabilization and calibration mechanism for Base mathematical reasoning.

A Full-Data Reference Results

Besides the five settings analyzed in the main body, the full-data and format-constrained reference results are useful for supplementary comparison. They should not be merged directly into the main result table because the training-file scope and logging coverage are not fully identical. Even so, they are valuable as appendix-level evidence because they reinforce the same broad pattern: in both model families, **full** training is stronger than **1-shot**, and a format-constrained reference for the Base family is much weaker than the standard reward-based path.

Table 8: Summary of representative full-data and format-constrained reference results

Model family	Setting	initial	final	best
Base	1-shot	0.3630	0.5753	0.5803
Base	full	0.3630	0.6053	0.6135
Base	format-constrained reference	0.3630	0.4490	0.4785
Base	1-shot entropy reference	0.3630	0.5698	0.5835
Instruct	1-shot	0.6860	0.6928	0.6985
Instruct	full	0.6860	0.7070	0.7070
Instruct	1-shot reference	0.6860	0.6893	0.7005
Instruct	full reference	0.6795	0.7118	0.7118

The table serves a narrow but useful purpose. It shows that the main paper narrative is not confined to the main result table alone: the Base family benefits more strongly in absolute score, **full** training tends to outperform **1-shot**, and a more heavily format-constrained reference remains substantially weaker. The corresponding visualization is already shown in Figure 1; the appendix therefore avoids repeating the same figures and uses this section only as a compact numerical reference. These results should be used as supplementary reference evidence rather than as part of the matched-protocol primary comparison.

B Metric Reading Guide

The main text deliberately reads score and behavior together rather than treating any single auxiliary diagnostic as decisive. That choice is especially important under a low-data RL regime, where the same scalar outcome can be reached through very different generation patterns. Table 9 therefore records how each diagnostic is interpreted in this paper and where its limits remain. The appendix-level guide is useful because several later figures compare settings in multiple metric spaces at once.

Table 9: How to read the behavioral metrics used in the paper

Metric	Typical favorable movement	What it reflects	Main caveat
Validation score	higher	Task-level correctness on held-out mathematical problems	Score alone does not reveal whether the model improved by better calibration, shorter search, or some other behavioral route.
Response length	lower	Less wandering and earlier commitment once the answer path is stabilized	Shorter outputs are only favorable when score is preserved or improved; otherwise they can reflect premature stopping.
Answer-tail confidence	higher	More confidence around the answer-bearing tail of the response	Tail confidence is informative only when paired with correctness, because confident wrong answers remain possible.
Low-confidence token ratio	lower	Less probability mass spent on highly uncertain token decisions	The absolute level depends on prompt difficulty, so cross-setting comparisons are more informative than any universal threshold.
Mean token entropy	lower	Sharper next-token distributions and less diffuse local uncertainty	Lower entropy can also appear under pathological collapse, which is why the random-reward case must be read jointly with score and EOS behavior.
Final EOS probability	higher	Clearer stopping behavior near the end of generation	A sharp EOS signal is only useful when it accompanies correct or improved answers rather than truncated failure.
Mean top-1 probability	higher	Stronger concentration on the most likely next token	High top-1 mass by itself does not distinguish correct confidence from confidently wrong behavior.

Three reading rules follow from Table 9. First, no single diagnostic is treated as a stand-alone success criterion apart from held-out score. Second, the most persuasive evidence comes from coordinated movement across metrics, such as shorter responses together with lower entropy and fewer low-confidence tokens. Third, the reward-corruption setting is informative precisely because it separates benign entropy contraction from pathological collapse: in that setting, entropy is not merely low or high in isolation, but coupled to vanishing score, saturated low-confidence mass, and near-zero EOS/top-1 probability.

C End-State Behavioral Summary for the Five Key Settings

The main body emphasizes trajectories because the path into the final regime is central to the paper’s interpretation. For quick appendix-level reference, however, it is also useful to collect the endpoint behavior of the five key settings in one place. Table 10 plays that

role and can be read as the numerical companion to the main trajectory figures.

Table 10: Final behavioral endpoints across the five key settings

Setting	score	length	tail conf.	low-conf.	entropy	EOS	top-1
Base n=8	0.5733	603.1	0.9331	0.0325	0.1839	0.9880	0.9354
Base n=1	0.5470	580.4	0.9003	0.0488	0.2811	0.9215	0.9227
Base weak-reg.	0.5747	578.6	0.9437	0.0294	0.1563	0.9738	0.9437
Base random-reward	0.0003	1192.4	0.0001	1.0000	11.5466	0.0009	0.0021
Instruct n=8	0.6873	470.3	0.9571	0.0187	0.1322	0.9926	0.9634

Note: Length is measured in tokens. The behavioral metrics are validation-time endpoints and should be interpreted jointly with score rather than as stand-alone optimization targets.

The table compresses the paper’s central comparison into a reader-friendly endpoint view. The successful Base settings cluster in a relatively narrow region of low entropy, low low-confidence mass, and high EOS/top-1 probability, while **Base random-reward** lies far outside that region. **Instruct n=8** finishes with the strongest endpoint profile overall, but its gain from initialization remains small. That contrast is exactly why the paper treats starting prior and gain scale as separate interpretive dimensions.

D Matched $n=1$ versus $n=8$: A Compact Side-by-Side View

The main text already argues that the $n=1$ group-size setting remains useful but does not fully match $n=8$. Figure 25 makes that comparison easier to inspect by placing the matched score curve beside five core behavioral trajectories. It is included in the appendix because the point is comparative rather than foundational: the paper’s main claim does not depend on this compact view, but the view helps the reader inspect the shape of the difference directly.

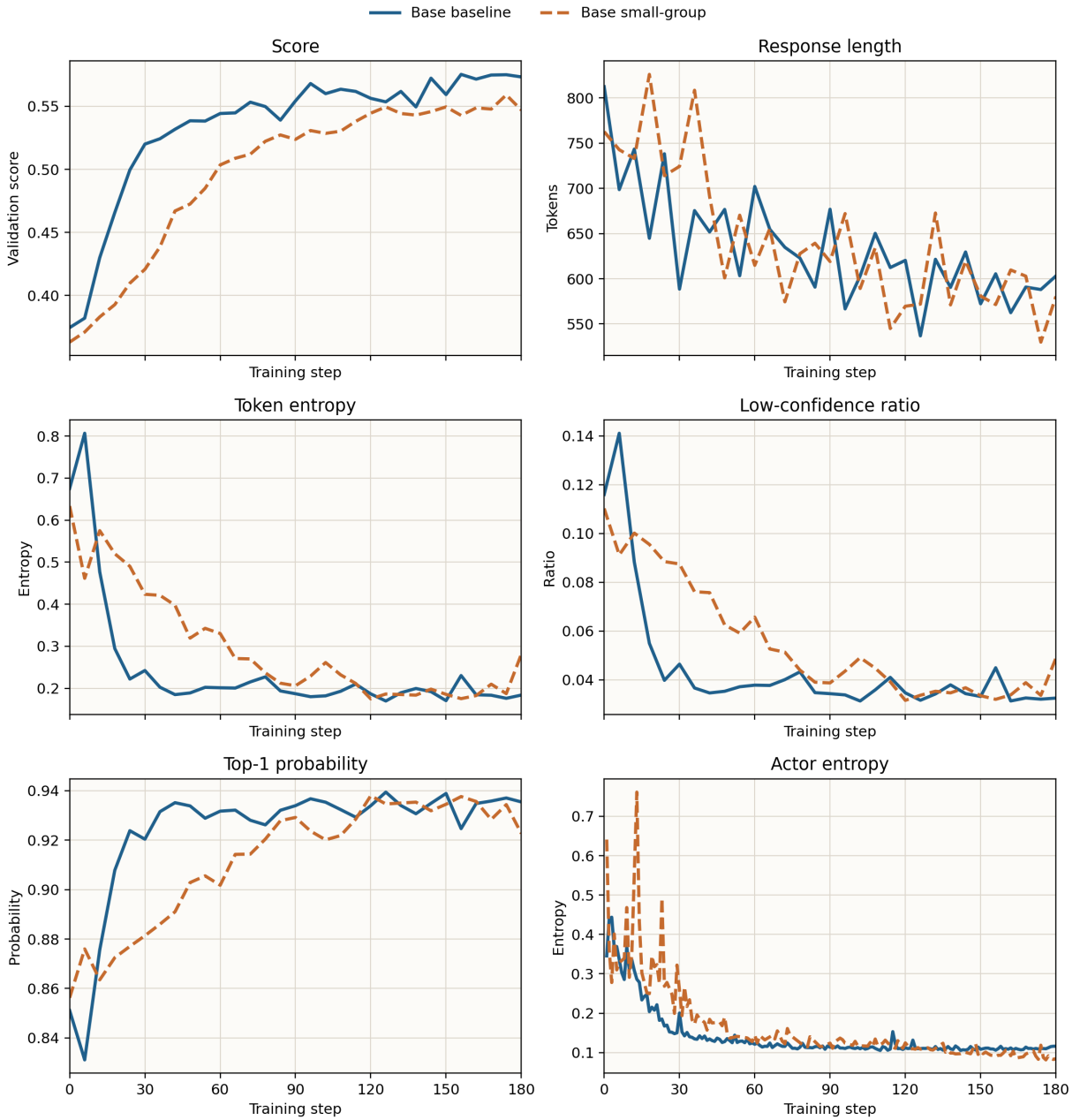


Figure 25: Matched comparison between Base $n=8$ and Base $n=1$ across score, response length, token entropy, low-confidence ratio, top-1 probability, and actor entropy.

The compact panel supports two observations that are easy to miss when each metric is read separately. First, **n=1** is not merely lower at the end; it also contracts more slowly in entropy and low-confidence mass during the early and middle phases of training. Second, the gap is not uniform across diagnostics. Final response length is not dramatically worse for **n=1**, but token entropy, low-confidence ratio, and top-1 probability remain consistently weaker, which fits the interpretation that the group-relative signal mainly strengthens distributional stabilization rather than simply shortening answers.

E Supplementary Comparison Among the Non-Collapse Settings

The non-collapse settings are easy to summarize too coarsely if one only says that they all converge to a “stable” region. Figure 26 expands that point by comparing the four non-collapse settings on four endpoint-sensitive metrics: token entropy, low-confidence ratio, final EOS probability, and top-1 probability. Unlike the main body’s broader trajectory coverage, this panel is meant as a compact appendix-level synthesis for readers who want to inspect how similar or dissimilar the successful regimes really are.

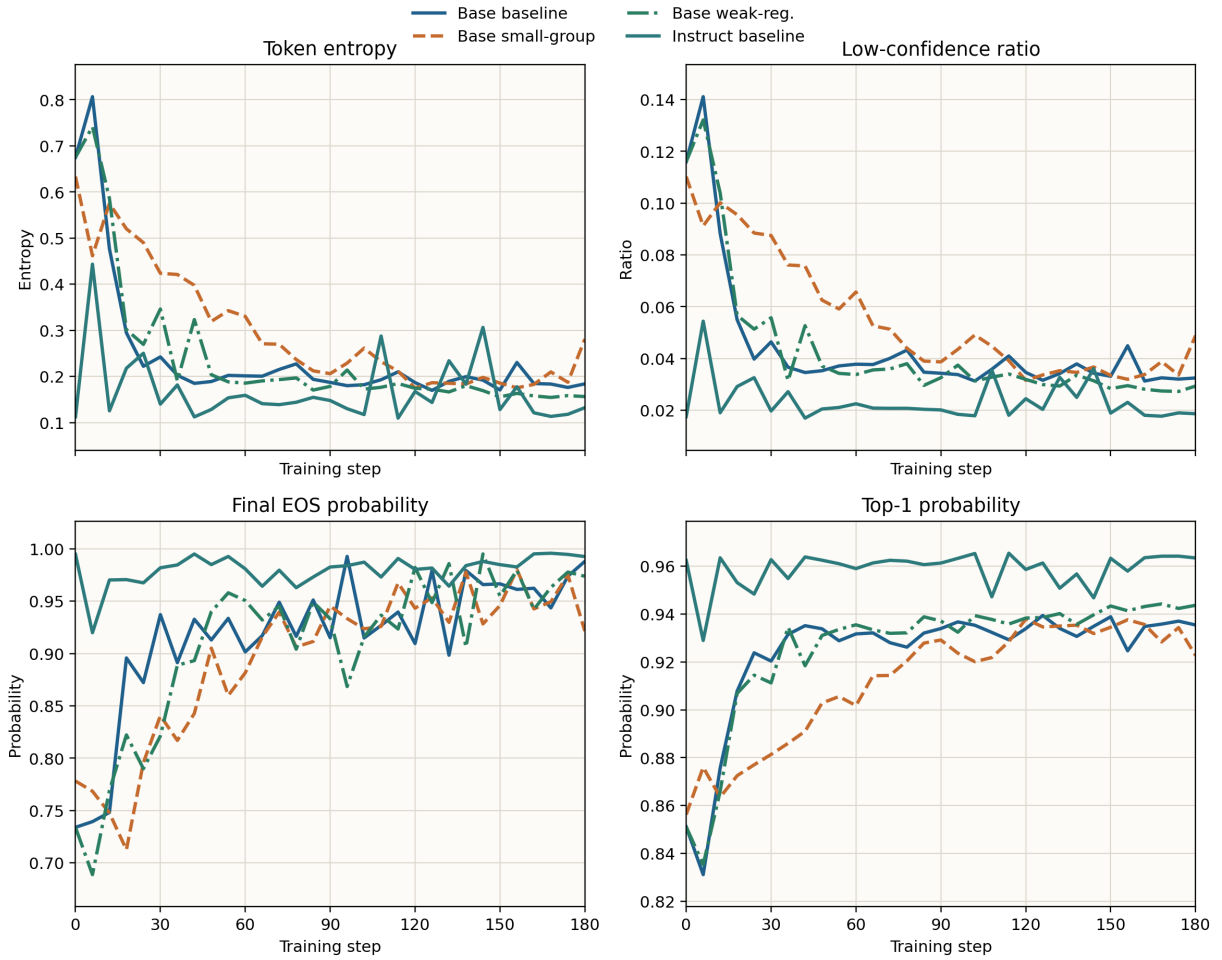


Figure 26: Supplementary comparison among the four non-collapse settings on token entropy, low-confidence ratio, final EOS probability, and top-1 probability.

This comparison reinforces a narrow but important point. The three successful Base settings do not converge to an identical endpoint distribution even when their final scores are close. The weak-regularization setting is the sharpest among them, **Base n=8** is slightly more moderate, and **Base n=1** remains visibly looser on entropy and low-confidence mass. The Instruct setting occupies the sharpest region overall, which is consistent with the paper’s main interpretation that it begins from a stronger prior and therefore has less

behavioral slack to correct.

F Correctness-Conditioned Entropy on the Traced Subset

One natural question is whether the token-level evidence only says that different settings behave differently, or whether correct and wrong responses also separate within a single setting. Figure 27 gives a small but concrete answer for the traced subset available in Base n=1. It plots final-step token entropy by correctness over eight traced MATH500 questions. The figure is intentionally placed in the appendix because the sample is too small to carry a main claim on its own, yet it is still useful as a directionally consistent piece of evidence.

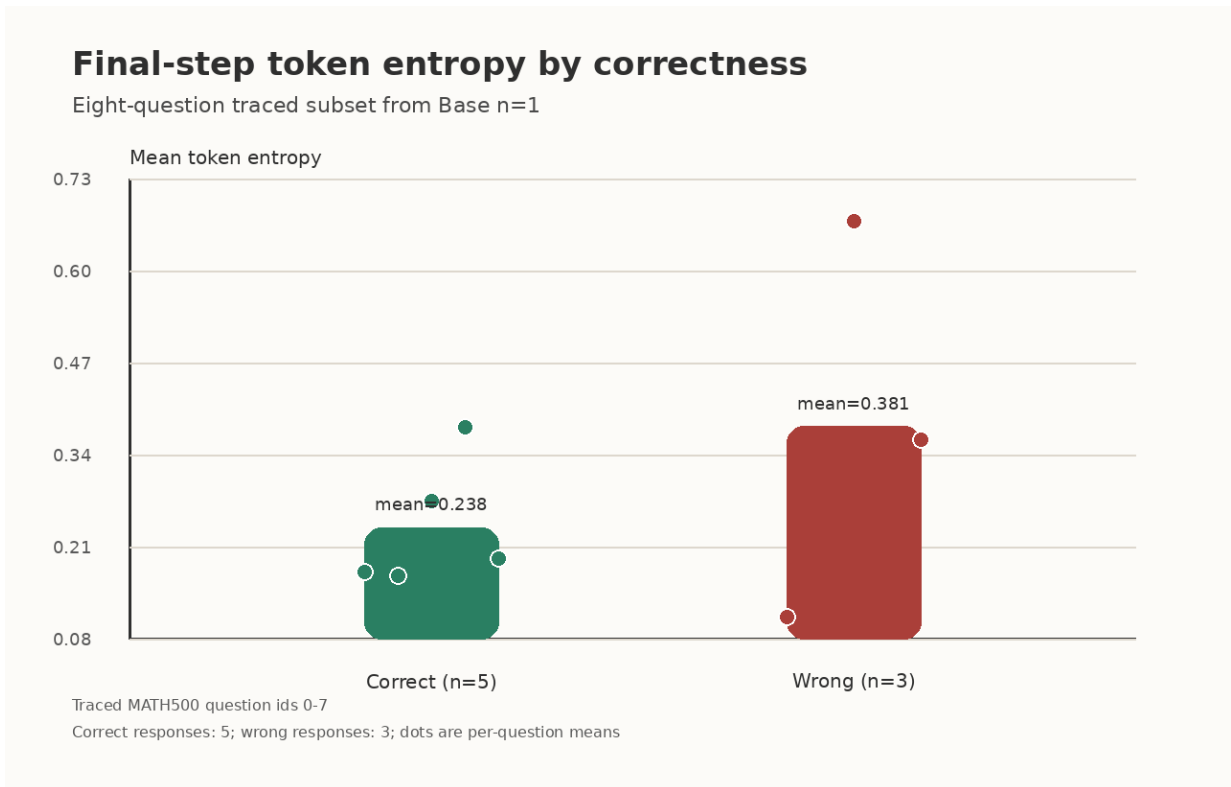


Figure 27: Final-step token entropy by correctness on the traced eight-question subset from Base n=1.

Within this traced subset, correct responses end with lower mean token entropy than wrong responses. That is exactly the ordering predicted by the calibration account, but the appendix keeps the evidential weight proportional to the sample size: the figure is best read as supportive micro-evidence rather than as a substitute for the aggregate metrics in the main body.

G Interpretation Boundaries for the Appendix Evidence

The appendix also clarifies two scope boundaries that matter when reading the supplementary material. First, the reward-corruption condition should be interpreted as an extreme wrong-signal control rather than a clean no-signal baseline. Its value is therefore causal rather than symmetric: it shows that meaningful reward integrity matters, but it does not by itself estimate what would happen under a perfectly neutral reward ablation. Second, the full-data references are included to position the low-data results under a larger-data empirical ceiling, not to replace the matched-protocol comparisons that anchor the main conclusions.

Those boundaries help explain why the appendix contains both richer comparison material and repeated caveats. The extra figures and tables are meant to improve readability, not to expand the strength of the core claims beyond what the matched main settings support. In that sense, the appendix is best understood as interpretive scaffolding: it makes the score, calibration, group-size, and reward-integrity story easier to inspect in detail while preserving the same evidential hierarchy as the main text.

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